

2

The basics of the system

This chapter provides a brief explanation of some important EPLAN principles, functions, and working methods, and uses a number of examples to illustrate selected facts and system settings. Important points in this chapter are the directory structure, data storage, user, workstation and project settings, notes on project and page properties, notes on particular dialog properties, handling of schemes, forms, plot frames, symbol libraries and an overview of my personal shortcut keys, optimized over the years.

■ 2.1 Five principles for working with EPLAN Electric P8



1st principle: Errors during project editing are allowed in EPLAN Electric P8.

In general, mistakes are allowed when working with EPLAN Electric P8. This basic principle of allowing mistakes is illustrated in the following example. A contactor may have two auxiliary contacts in the schematic, with both of them initially having the same connection point designation. Things that are not physically possible are initially “allowed” by EPLAN while working on a project. The user is not slowed down by “irritating” errors or messages during editing. This type of error naturally appears in message management, but only as a message entry, depending on the setting within the check run selected. This entry initially has no further consequences. When project editing has progressed far enough, or is finished, EPLAN can perform certain project checks. Erroneous entries such as those described above will be listed in the project, if they do not already exist in message management.

Of course this error must be fixed in order to have a correct practical reference. However, this is not **compulsory**. EPLAN allows the user to decide whether a project is error free (no message) or not and which priority a message (error, warning, or note) should have. It is also possible to prevent such errors. With the **Prevent errors** check option, the above approach would not be possible. But this is a user-settable setting.



2nd principle: In EPLAN Electric P8, what is selected is what is edited.

The following example clarifies this principle. If I select three texts on a page and start the translation function, then exactly these three texts will be translated. If I select this page in the page navigator, then the translation function will translate all texts on the entire page based on their settings.



3rd principle: EPLAN Electric P8 stores data and any references online.

EPLAN is an online system. All references and device data are constantly (i.e. online) updated. For performance reasons, these are only updated on request via a few specific actions. EPLAN performs the rest completely independently.

A typical example of this is the editing of a page followed by a page change. Here you need to manually start any required updating of the connections. There is of course a setting that allows EPLAN to do this type of connection updating automatically. However, this can negatively affect project performance.

In my opinion, constant (online) updating of connections is not really necessary because relevant actions such as graphical project reports or automated procedures such as device numbering automatically update the connections before the actual action is performed.



4th principle: EPLAN Electric P8 can be operated using a graphical approach.

This means that the devices (symbols) can first be placed in the schematic and subsequently be assigned the parts, including the associated function definitions. This is not compulsory, and you have a completely free hand when editing a project.



5th principle: EPLAN Electric P8 can also be operated using an object-oriented approach.

This means that external motor lists or other component lists can be read into the system as device lists, and the project can be started from this end.



NOTE: Principles 4 and 5 can be combined with each other. There are absolutely no limitations when working with EPLAN Electric P8.

2.2 Directory structure, storage locations

EPLAN can use any desired directory structure. EPLAN allows the user a free choice here. Data, such as project or master data, can therefore be easily integrated into an existing company data storage structure.

EPLAN recommends installing/running the program files locally, and only storing the data on the network. I agree with this recommendation. Any further structuring or organization of the remaining master data is the responsibility of the user.

By default, EPLAN creates the program directory with the following sub-directories:

- **BIN:** contains the program modules and, in additional subdirectories, the language files (e.g. en-US for U.S. English).
- **CFG:** contains configuration files for the users, company, workstation and projects.
- **P_ID:** is only of interest to PPE users.

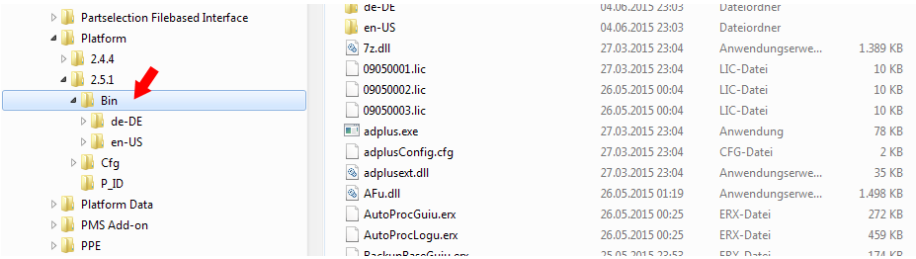


Fig. 2.1 Program directory with subdirectory

Similarly, during installation EPLAN also creates certain default directories for particular system data and other data.

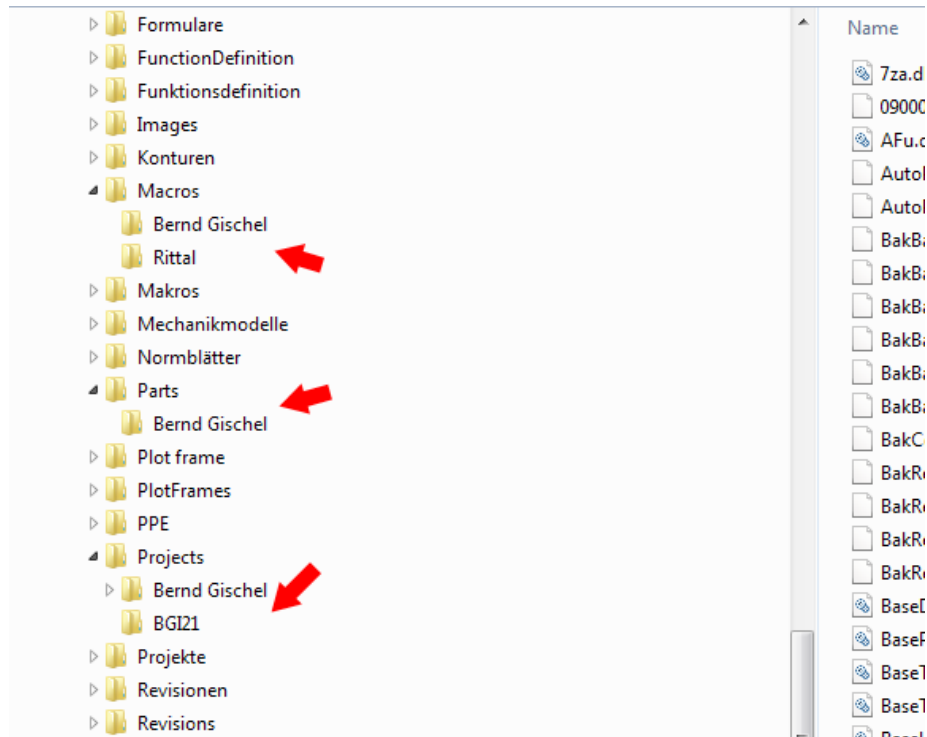


Fig. 2.2
Default customer
directories

For example, EPLAN creates a main directory called *Parts* with the subdirectory *Company code*. EPLAN gets this code from the information entered during installation – in this case *Bernd Gischel*.

The following *directories* are initially created, among other things, as default directories and filled with the relevant data during installation (the most important directories are listed):

- *Part*: contains the parts databases (Microsoft Access *.mdb format) and the configuration files for importing and exporting parts
- *Images*: contains all images, such as images for parts data
- *Documents*: contains documents such as PDF documents or Excel tables
- *Dxf_Dwg*: contains CAD drawings in DXF or DWG format
- *Forms*: contains all forms (own system master data)
- *Function definition*: contains the files for the function definitions
- *Macros*: contains all macros, such as window macros (*.ema), symbol macros (*.ems), page macros (*.emp) or also 3D macros for EPLAN Pro Panel
- *Mechanical models*: contains mechanical data, such as 3D models
- *Plot frames*: contains all plot frames (own system master data)
- *Projects*: default directory for projects
- *Schemes*: contains preconfigured or user-created schemes, for example personal filter or sorting settings.

- *Scripts*: contains the corresponding *.cs or *.vb script files
- *Symbols*: contains all symbol libraries (own system master data)
- *Translation*: contains the dictionaries (translation databases in Microsoft Access format, *.mdb)
- *Management*: standardly contains the project database into which all projects are imported from project management and other management databases such as the rights management database
- *Templates*: contains templates, basic projects and exchange files for exporting project data (labels)
- *Xml*: contains XML files

In addition, during installation EPLAN stores a directory with the original EPLAN master data for subsequent synchronization with its own master data. This directory can be chosen freely during installation.

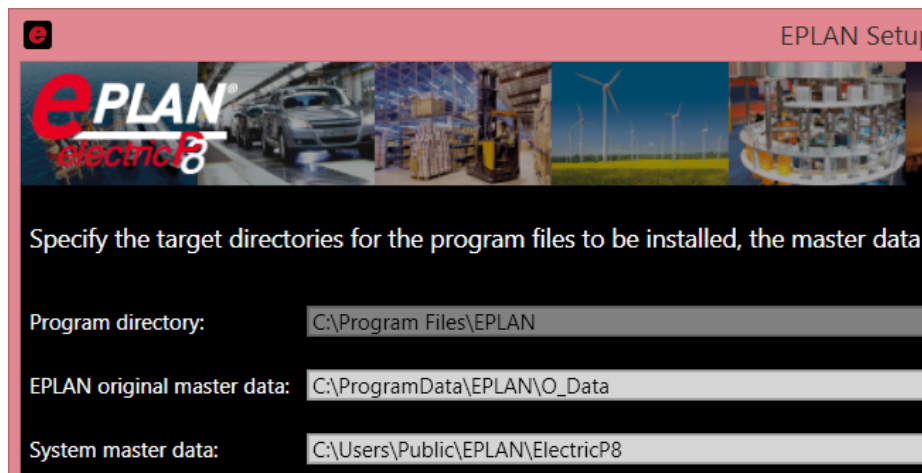


Fig. 2.3
Example: directory for
EPLAN original master
data

These are the standard EPLAN defaults. EPLAN users can, of course, change these directory structures to suit their needs. For practical reasons, this should be done after the initial program start and before starting project management for the first time.

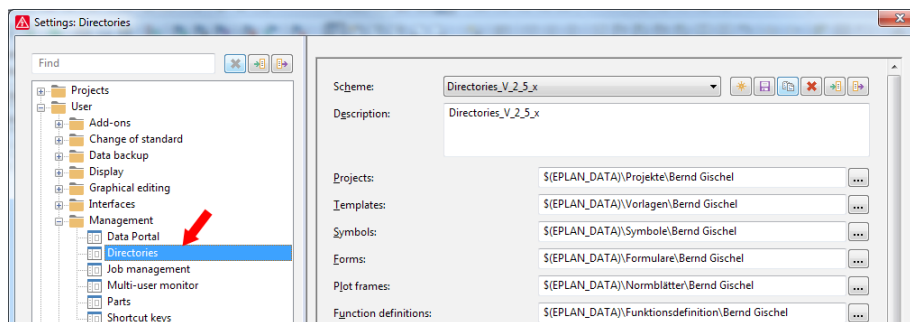


Fig. 2.4
Directory settings

User /
Management /
Directories

These directories can be viewed and modified under **OPTIONS/SETTINGS/USER/MANAGEMENT/DIRECTORIES**. A useful feature is that separate schemes can be created and configured for different directory settings here.



TIP: It is also possible to start EPLAN Electric P8 with an appropriate command line parameter to automatically set a particular directory structure. Example: "J:\EPLANP8x64\Platform\2.5.2\BIN\Eplan.exe" /Variant: "Electric P8" /PathsScheme:OwnSchemeName (without prefixes and suffixes). This makes it possible to use customer directories with different system master data selectively.

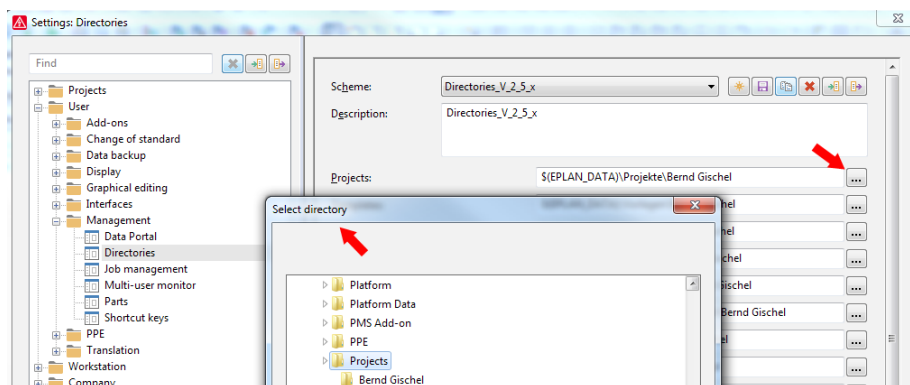


Fig. 2.5 Selecting other directories

Clicking the  button causes EPLAN to display the **Browse For Folder** dialog. You can select a different directory here, or use the **CREATE FOLDER** button to create a folder at the desired storage location.



NOTE: These changed settings will not be used until EPLAN is restarted.

■ 2.3 Settings – General

To put it simply, a project contains all the relevant data that it uses and requires. This means that a project is independent of the general system master data because EPLAN stores all required and used data within the project automatically upon first-time use.

The advantage of this method of “storage” should not be underestimated because even after several years this type of project can still be opened and edited with the original master data such as plot frames, symbols, etc. This is also true, for example, when the graphics of a company plot frame changes (e.g. because the company logo has been revised).

If the user wants, this behavior can be configured in EPLAN via settings (parameters). There are four basic setting areas as described below.

The settings are accessed via the **OPTIONS/SETTINGS** menu item. Storing this variety of project-related data naturally increases the data volume of the project.



An empty project has an average size of approx. 4 MB to 5 MB.

To make finding settings faster, EPLAN has integrated a search function with the **Settings** dialog.

The search itself is quite simple. Simply enter the desired term (the property you are looking for), such as “minimum”, and EPLAN will instantly list suitable terms.

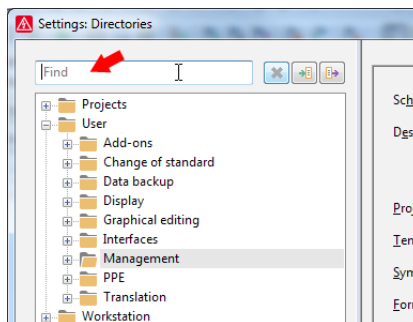


Fig. 2.6
Search field

Upper- or lowercase does not matter here. EPLAN simply looks for the text. It may be part of a word, and the word does not even have to begin with it.

Operations (AND/OR) and placeholders like * or ? do not work with this search function. Nor are numbers found, but only pure texts contained in the settings and/or their dialogs.

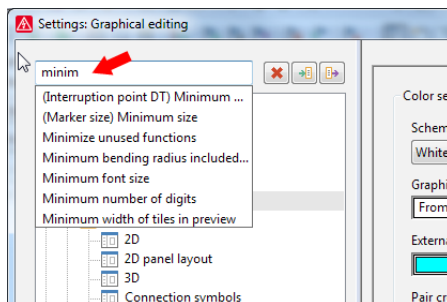


Fig. 2.7
Example: input of search term

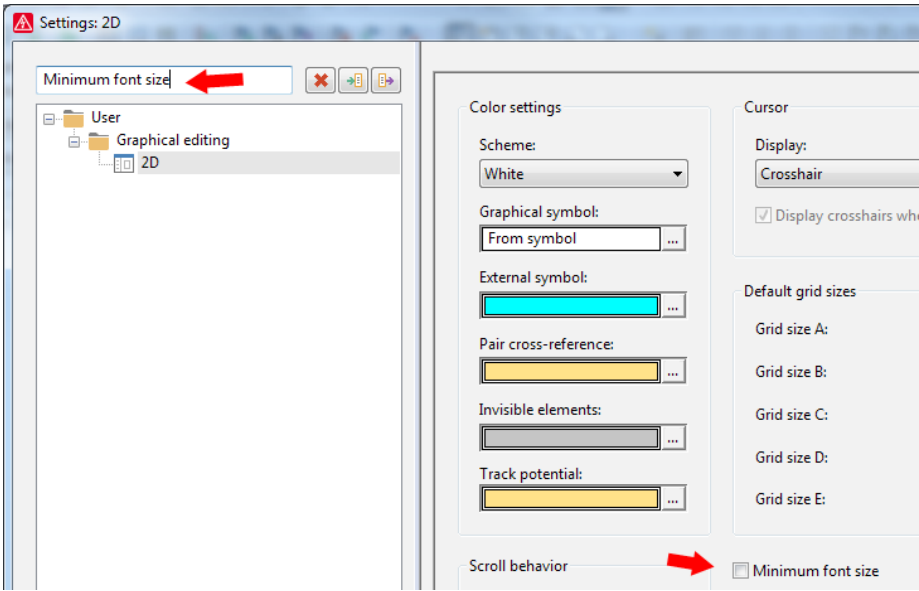
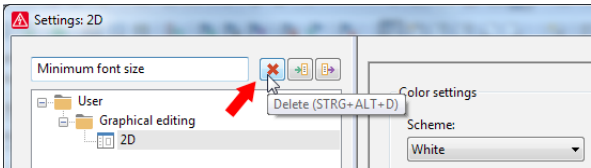


Fig. 2.8 Search results, minimum font size

If you click on the searched term, EPLAN opens the corresponding setting, and all other setting options in the tree structure are hidden. To display them again, the search must be 'emptied' first. This is done by clicking the **DELETE** button.

The following sections provide a brief overview of the settings. Many of these settings are explained in more detail in other chapters where they are actually used and can be more clearly explained.

Fig. 2.9
Search results,
minimum font size



2.3.1 Settings – Project

All project properties (and only these) are defined under **SETTINGS/PROJECTS [PROJECT NAME]**. The settings in the *Projects* node are available only if at least one project is open in the **page navigator**.

In the project settings you can, for example, define whether the project master data should be automatically synchronized with the system master data when the project is opened. You should always carefully consider the use of this setting, and many other project settings (i.e. “automation” when a project is opened), because changing these set-

tings can have wide-reaching effects on the project to be opened. EPLAN has no “undo” button at this point, and if no data backups exist the old data is irretrievably lost.

Some important project properties, for example, are located under **OPTIONS/SETTINGS/PROJECTS/PROJECT NAME/MANAGEMENT/PAGES** and under the entry for the *Default plot frame*. This entry for the global plot frame applies to all pages, even any report pages, if no other plot frame has been assigned manually via the page properties. Symbol libraries are located under **OPTIONS/SETTINGS/PROJECTS/PROJECT NAME/ MANAGEMENT/SYMBOL LIBRARIES**. After the settings are selected and entered, they are also stored with the current state in the project.

There are numerous settings – too many to include in the scope of this book. For this reason, only a number of important properties are listed and explained.

2.3.1.1 Setting Projects [project name]/Reports

This area affects the output of graphical reports (see chapter 6, “Reports”, for further explanations).

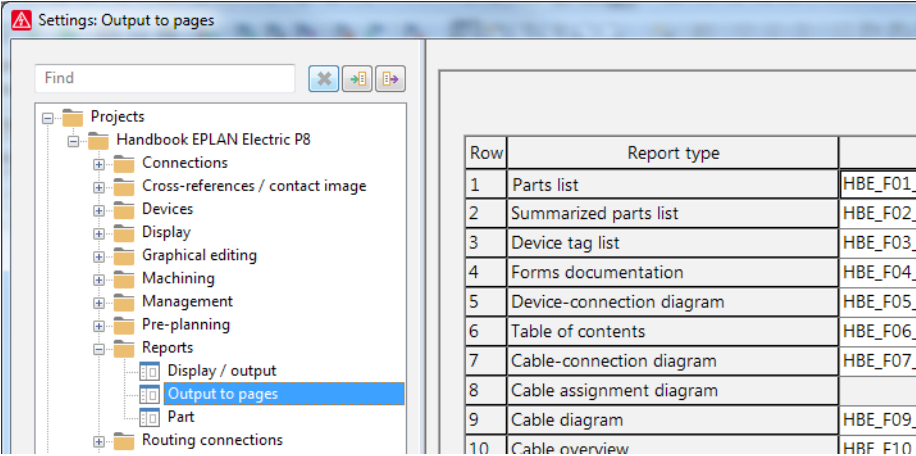


Fig. 2.10 Settings for reports

- **Display/Output:** Settings that affect the output or display of project data in reports (forms)
- **Part:** Settings that affect the output of parts data and their behavior in reports and their display in certain navigators
- **Output to pages:** Project-wide settings for reports (forms) for the different report types in a project

2.3.1.2 Setting Projects [project name]/Devices

The device area is where you define different settings such as numbering schemes or syntax checks for devices.

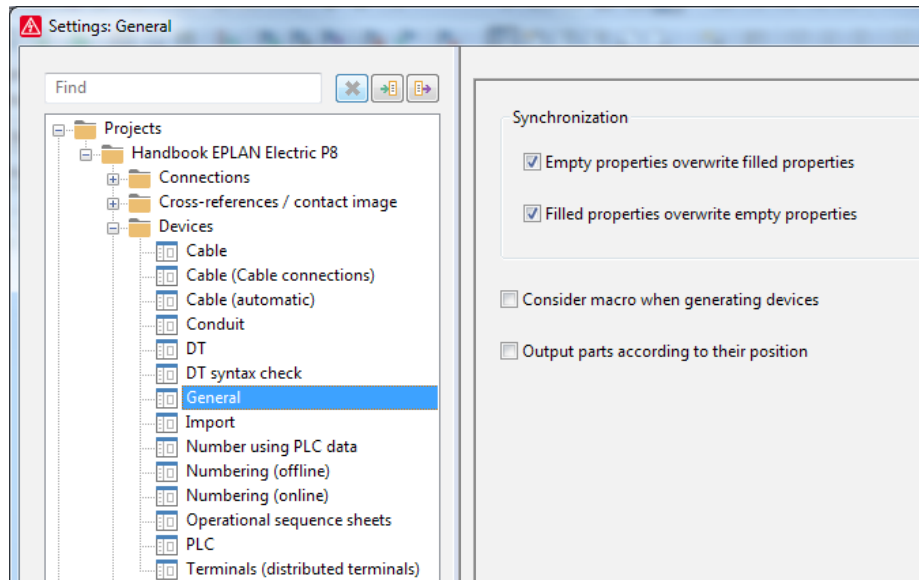


Fig. 2.11 Settings for devices

- **Sequence sheets:** Contains defaults for the symbols that are placed in sequence sheets (such as simple sequences (sequence chains) for a graphical listing of plant or machine processes according to the GRAFCET standard DIN EN 60848).
- **General:** Contains settings for the synchronization of functions, determining whether empty properties here may overwrite filled properties or vice versa. This is also where you find settings to specify whether a macro stored at the part is to be considered when generating new devices (such as in a navigator, via the New Device menu entry).
- **DT:** Contains general settings, including the parameter for *conversion to uppercase* for devices. If this parameter is changed, uppercase letters are not automatically converted to lowercase letters or vice versa. It is only possible to work with either uppercase letters (parameter activated) or with uppercase and lowercase (parameter not activated). Generally, EPLAN initially always uses uppercase letters for identifiers. This setting can also be used to define the devices that automatically receive a prefix on insertion.
- **DT syntax check:** These settings define the possible *special characters* for structure identifiers and for the input of device tags (electrical engineering/fluid trade)
- **Import:** EPLAN allows for the import of device data. Schemes are used to define the allocation, that is, which external data belong to which EPLAN property.
- **Cables:** Default settings for inserting cable definition lines (cables), shields, and their connection definition points (i. e. which connection definition point graphical symbol is the first choice). A general standard cable can also be specified as a default for the project.
- **Cables (cable connections):** These settings allow for default settings; such as defining the symbols that EPLAN should use when inserting a connection definition in connec-

tion with cable definition lines and when inserting shields. These settings apply to multi-line and single-line representation types.

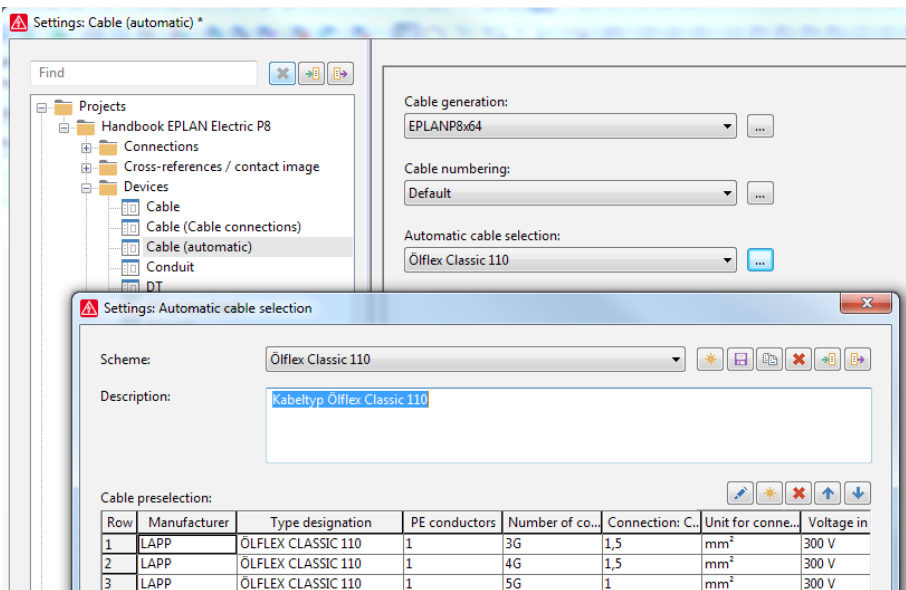


Fig. 2.12 Automatic cable selection (cable node (automatic))

- **Cables (automatic):** Contains the settings for automatic cable generation, cable selection, and cable numbering. All settings are defined via filter schemes. You can use pre-defined schemes or define your own schemes.
- **Terminals (distributed terminals):** Contains defaults for working with distributed terminals, such as: What function definition should be used for distributed terminals or at what point should terminals automatically become distributed terminals when placed (number of connection points)? This also contains the option for optimizing jumpers.
- **Numbering with PLC data:** Default values of a scheme for numbering connected devices with PLC data. PLC numbering can be used to number the following devices (among others): Terminals, pins, and general devices.
- **Numbering (offline):** Settings for a scheme for numbering devices offline (subsequently). Offline numbering is used to subsequently give a schematic a different DT layout. For example, the devices were first numbered with the default [identifier counter]. However, the devices must now be numbered according to the scheme [page identifier column]. The default for the scheme is set here. However, this does not have to be used. The scheme for offline numbering can be changed at any later time.
- **Numbering (online):** Contains the defaults for a numbering scheme for online device assignment. Such schemes are applied immediately when you create a schematic or insert symbols, macros or copy operations. This setting also defines the identifier set used for the project or how identifiers are to be handled when inserting symbols or macros.

- **Conduit:** Contains the settings for which connection definition point symbol should be placed when creating the EPLAN Fluid add-on's conduit definition lines (such as a tube or pipe connection).
Note: A conduit definition line is inserted in the same way as a cable definition line.
- **PLC:** The defaults for such output as PLC assignment lists, the settings for the extended path function text import (extension of the path in order to import also "adjacent" function texts "on the left or at the top"), and for the output format.

2.3.1.3 Setting Projects [project name]/Display

This setting is used to change the display of project structures in the navigators as well as in the page navigator. The *Display* node also contains the display or format of numbers, time and date.

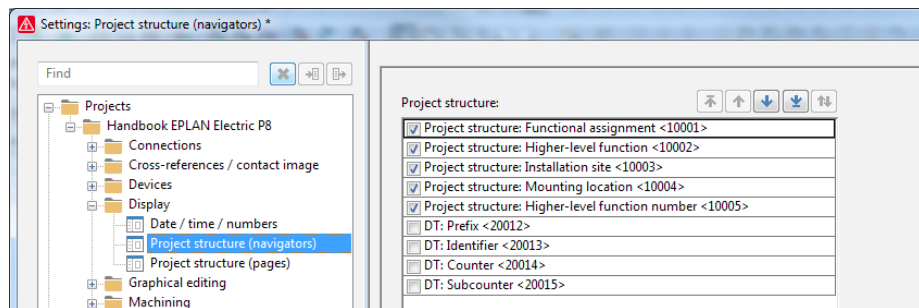


Fig. 2.13 Setting the project structure and format display

- **Date/Time/Numbers:** Settings for selecting the display format of the date, time and numbers in accordance with the operating system or a project-specific format (for example to set the date or time format to match typical foreign displays).
- **Project structure (navigators):** These settings allow you to change the display sequence of structure identifiers and/or of the DT in the navigators (except for the page navigator). However, this relates only to the display in the various navigators. Devices are not changed. Here, you can also define whether the substructures of the structure of a device should be represented as ungrouped or grouped.

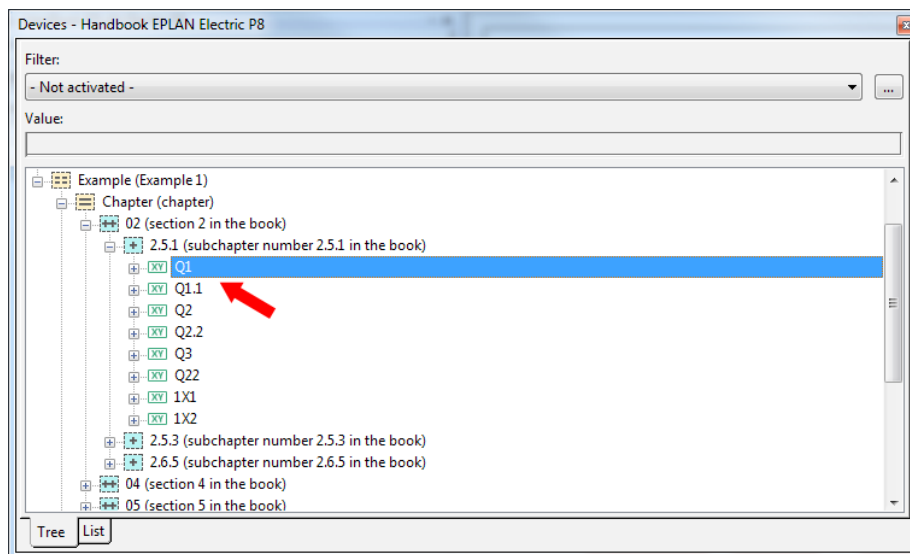


Fig. 2.14 Grouped structure of the device (display only)

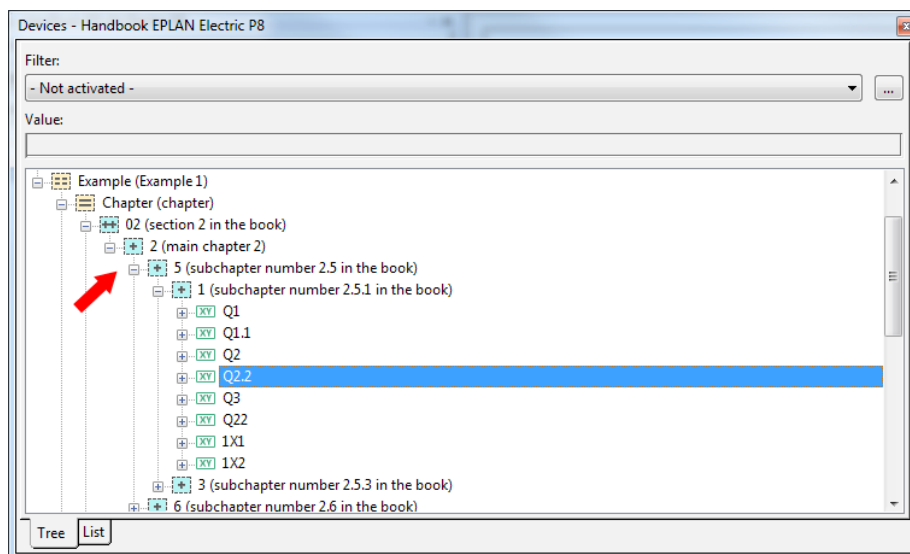


Fig. 2.15 Ungrouped structure of the device (display only)

- **Project structure (pages):** These settings are used to adjust the display sequence, especially in the page navigator (Page/Navigator menu) or via the F12 function key. Like in the previous node, there is a setting here to represent the structures as ungrouped.



NOTE: These settings change only the representation or display of the project structure or devices in the navigators. The page structure and/or the device tag structures are not changed.

2.3.1.4 Setting Projects [project name]/Graphical editing

This section covers selected general settings for graphical editing.

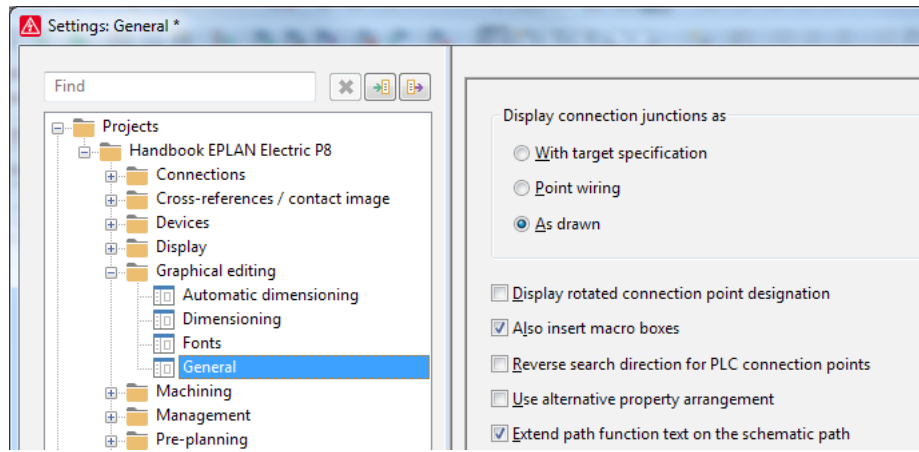


Fig. 2.16 Settings for graphical editing

- **General:** Settings such as the default values for representing connection junctions are set here, i.e. whether the display should be target-oriented, a point, or shown as drawn. However, the setting depends on the **SETTINGS/USER/GRAPHICAL EDITING/CONNECTION SYMBOLS** user setting and especially the *Connection symbols with prompt* parameter.
- **With target specification:** This default means, for example, that a T-node is drawn as target wiring. This T-node is also first placed and can then subsequently be changed to point wiring or a different target. This parameter has the effect that the representation remains as target wiring, regardless of how the T-node is internally set.
- **Point wiring:** This default allows T-nodes (junctions) to be entered with the visual display of point wiring. Internally, this type of junction can also be set for target wiring.
- **As drawn:** This means that a T-node, for example, is generally placed as target wiring. It is not possible to change this while placing (inserting) in the schematic. After the T-node has been placed, the **T-node [direction]** dialog can be called up by double-clicking and the targets can be changed. It is possible to change the display setting of this T-node from target wiring to point wiring. This representation is then also shown in the schematic. This setting thus allows a mixed representation of target and point wiring. With these parameters, it is generally possible to switch between the point wiring and target wiring representations. This means that if all T-nodes (junctions) were drawn

with the point representation, the target wiring parameter allows the representation to be changed to target wiring.

Insertion of T-nodes (junctions) is controlled at a higher level by the **SETTINGS/USER/GRAPHICAL EDITING/CONNECTION SYMBOLS** parameter using the *Connection symbols with prompt* parameter.

If this query parameter is not switched on, the T-node (junction) is first placed and can be changed later. If the dialog is switched on, it is displayed before the T-node (junction) is placed in order to set the target tracking or point/target wiring.

Aside from the settings for the connection symbols, there are other additional general settings contained in this node.

▪ **Display rotated connection point designation:**

When this setting is activated, the connection point designations are displayed rotated by 90 degrees. This setting also applies to plug and terminal designations.

▪ **Also insert macro boxes:** If this setting is activated, any macro boxes present on the macro will also be inserted on the page.

▪ **Reverse search sequence for PLC connection points:** If this setting is activated, EPLAN will look for the path function texts

against the connection point direction of the symbols. Usually, for example, EPLAN would run a downward search for the associated path function text at a PLC input placed at the bottom of the page and connected upward, and only if EPLAN finds no path function text will it continue the search upward. If this setting is then activated, EPLAN will look for a path function text immediately in the connection point direction.

▪ **Use alternative property arrangement:** Setting to achieve a representation according to the GOST standard. When this setting is activated, the properties to be displayed are represented on the right or at the top right of the symbol – in contrast to the ‘regular’ left-side representation.

▪ **Extend path function text on the schematic path:** If this setting is activated, the insertion point of a path function text does not have to be directly in the path of the device in order to be applied. It is sufficient if the path function text is in a path (the path and/or column width to be searched is obtained from the selected plot frame).

▪ **Draw structure boxes with recess:** This setting allows for EPLAN structure boxes to be drawn automatically with a recess. Instead of the usual rectangle, for example, a recess is drawn in which the DT is placed, depending on the property arrangement.

Fig. 2.17
Additional general settings

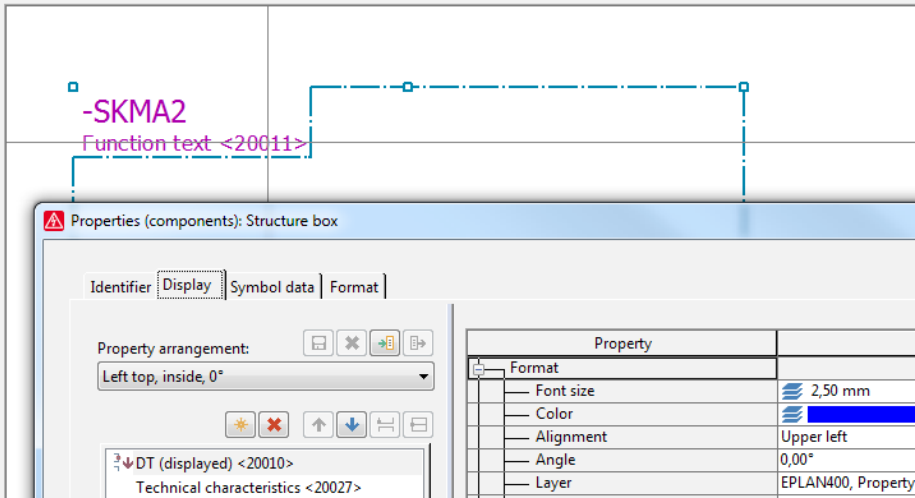


Fig. 2.18
Structure box with
recess

- **Import property arrangements from the project:** If parts with configured, user-defined property arrangements are copied from a project, they are compared when being inserted into another project. Depending on the setting of this parameter, they are applied to the new project (including the given property arrangement name), or you do not overwrite the existing property arrangements, in which case you will then see the property arrangement name “user-defined” instead of the name you used in the source project.
- **Text box size:** Default width x height settings for the size of the text box (can be selected individually in the **Properties (Texts)/Format Tab** dialog).

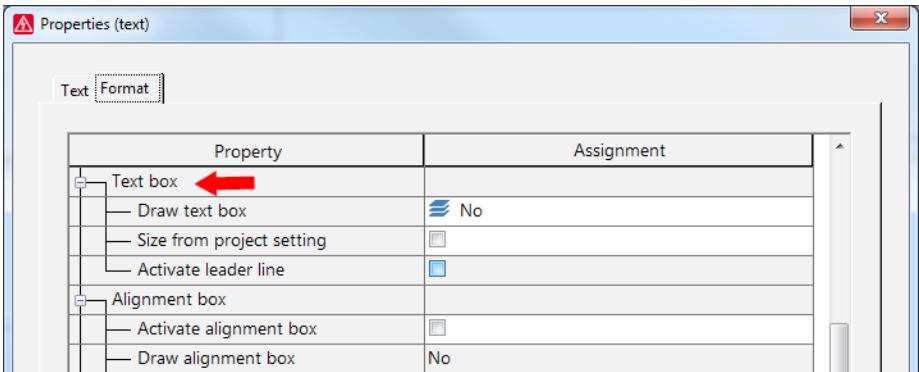


Fig. 2.19
Setting of text box in
the Properties (Text)
dialog

- **Model view:** Settings for the background colors of the model views of panel layouts.
- **Minimum font size for checks:** This setting is used to configure the check value (Other message class, message number 022027; Text too small message text) of the font size that must not be undershot in the project.
- **Dimensioning:** Format setting for entering dimensions, such as font or dimension line termination. Changing this setting has no effect on existing dimensions, and only affects

dimensions that are subsequently inserted. The number of decimal places is defined in the USER > DISPLAY > DISPLAYED UNITS OF MEASURE setting.

- **Fonts:** Defines the fonts that should be used in the project (up to ten fonts). These settings have priority over the font settings in the company settings (*Company/Graphical editing/Fonts*).



NOTE: The settings in the “Automatic Dimensioning” node do not apply to the 2D panel layout. This is why they will not be explained further here.

2.3.1.5 Setting Projects [project name]/Cross-references/contact image

These settings offer many parameters defining the representation of references, such as for interruption points, functions, such as motor overload switches, contactors, as well as general areas (for example the representation of references between different page and report types and the symbols to be used) and the display itself (the options for the overview cross-references between the types multi-line, single-line and the overviews are set here).

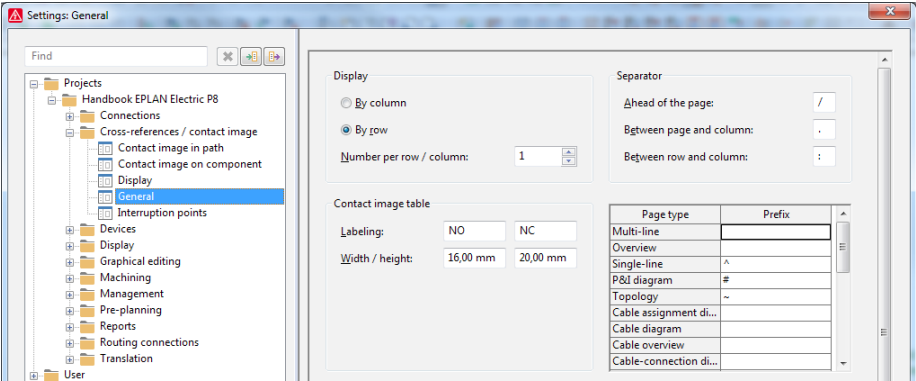


Fig. 2.20 Cross-references/contact image settings

The device settings always have priority.

If no changes are made to the device settings, for example to the cross-reference representation, the default settings are taken from these parameters (depending on the symbol properties).

- **Interruption points:** Default settings for displaying cross-references such as by column or row, the separator between the name of the interruption point and the cross-reference, the representation of cross-references, as well as settings for the representation of the target display at the interruption points.
- **General:** Settings for displaying general cross-references, how and which separators are placed between pages and columns and if the cross-reference is enclosed in brackets

- **Display:** Global project setting defining between which representation types (multi-line, single-line overview, topology, etc.) device cross-references should be displayed. These settings can be configured for both directions.
- **Contact image on component:** Default setting for displaying cross-references at components, such as motor overload switches, circuit breakers, as well as the default setting for a form to be embedded for the contact images. The form type is the distributed device list *.f45.
- **Contact image in path:** Default setting for displaying, for example, cross-references of the unfolds (contact image) of contactor coils. Just like with the previous setting, here, too, a global form for the contact image can be configured instead of the default settings.



-Q1			
3RT1015-1HB41 [Siemens]			
3RT1011-2GA04 [Siemens]			
1	NC contact	2	/1,6
3	NC contact	4	
5	NC contact	6	
13	NC contact	14	/1,7
51	NO contact	52	
61	NO contact	62	
71	NO contact	72	
81	NO contact	82	/1,8
81	NO contact	82	/2,8



Fig. 2.21
Example: contact
image with and without
a form

2.3.1.6 Setting Projects [project name]/Connections

The default values for connection properties are defined in this node.

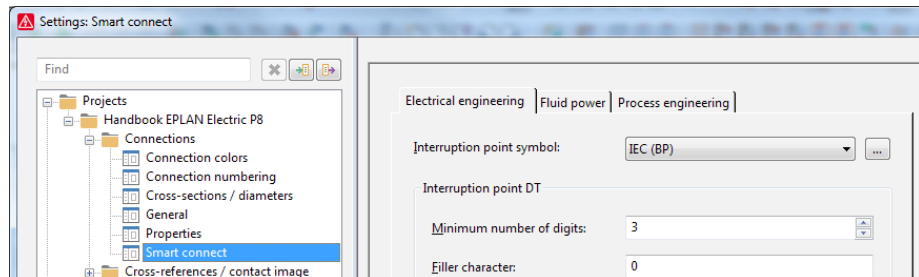


Fig. 2.22 Connection settings

- **General:** Some settings for determining the source and target
- **Properties:** Project-wide default settings for specific properties of a general connection, such as a global default value (e.g. 0.75), the unit (mm²), or also a global default value for the color of a connection. These settings can be configured for the various trades (Electrical engineering, Hydraulics, Pneumatics, and many others).
- **Cross-sections/diameters:** Extendable table with a project's default settings for cross-sections and diameter values.
- **Smart connect:** Defaults for settings such as using a special interruption point symbol and for interruption point designations of interruption points that EPLAN automatically generates when using smart connect.
- **Connection colors:** Editable, extendable table of color codes, color names and colors to be used in a layout space.
- **Connection numbering:** Defines the scheme used for numbering connections. You can use predefined schemes or your own scheme.

2.3.1.7 Setting Projects [project name]/Routing connection

This is where you enter default values for connections, their routing and associated terminations such as double-ended sleeves. The data/settings are needed during export.

This is also where you will find the settings for importing (scheme) connections and their information, such as source and target DT, conductor color and cross-section.

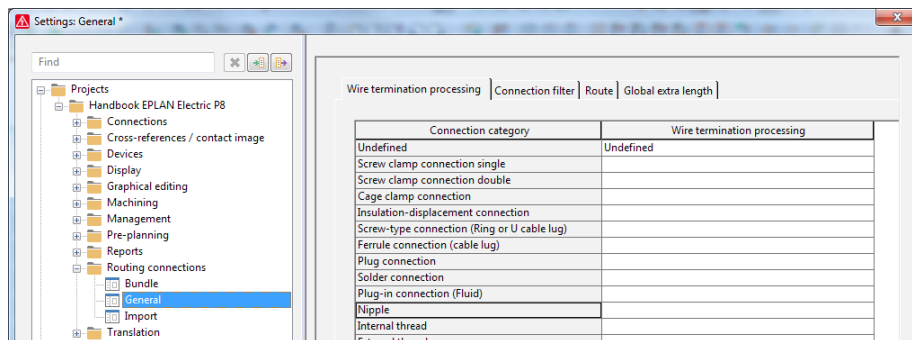


Fig. 2.23 Settings for routing connections

2.3.1.8 Setting Projects [project name]/Management

This includes default settings defining how EPLAN should handle, for instance, master data or particular modules in the project.

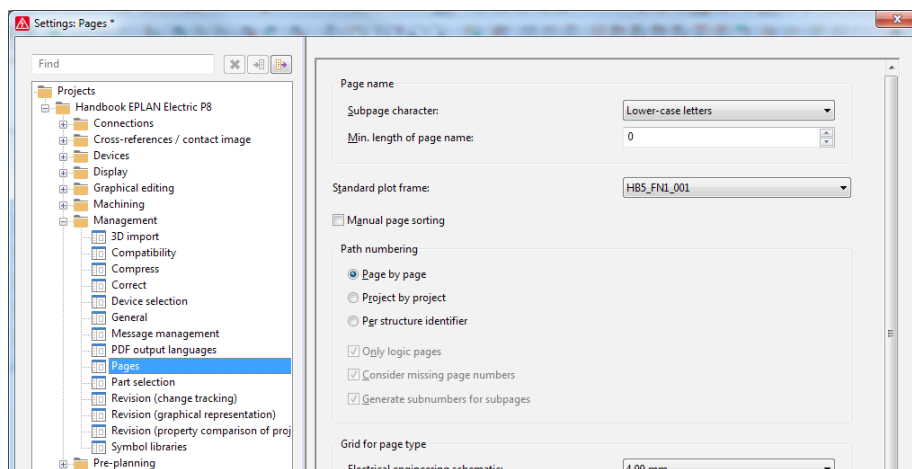


Fig. 2.24 Settings for managing the project

- **3D import:** Default setting of a function definition when importing 3D data
- **General:** This setting contains three very important parameters. The *Synchronize project master data when opening* parameter means that when the project is opened, the project master data, e.g. a terminal diagram form or a symbol library, is synchronized

with the system master data and possibly updated. The *Synchronize plot frames* setting synchronizes plot frames immediately after the project start. The last setting, *Synchronize stored parts when opening*, means that parts are automatically synchronized with the master data.



NOTE: Forms are only synchronized when, in the Projects [project name]/ Reports/Output to page setting, the Synchronize option for the corresponding forms has also been set.

Before EPLAN synchronizes the master data, it displays a security message that must be confirmed by clicking **YES**. Only then is the project master data updated, and only when it is older than the system master data.

- **Part selection:** Preselection of parts (filter schemes, for example, can be used to display only particular types of contactor in the parts selection), and also which parts database EPLAN should use
- **Device selection:** In addition to part selection, EPLAN also has more effective device selection. Whereas the part selection is independent of the function definitions used in the schematic, device selection occurs based only on the function definitions belonging to the devices. A preselection of how EPLAN should handle the device selection is defined here: For example, are existing function data to be used and if so, which ones, etc.?
- **Compatibility:** Settings for the 'downward compatibility' of legacy projects and/or legacy functions. These settings should ideally no longer be used when editing a project.
- **Compress:** Default scheme used for compressing the project
- **Correct:** Default setting of a scheme to correct project data, such as terminals, plugs, cables and connections
- **Message management:** An important setting that defines the extent and type of error checking for the current project. Project checks are not essential and do not necessarily have to be performed. However, they do uncover possible editing or data entry errors during project editing and list the problem areas in message management. This setting defines the error checking and the extent of each check for this project. The number of project check messages can also be limited here. This scheme is stored in the project the first time it is used, but is not automatically synchronized with a newer version of this check run. This is important to remember when it is used.
- **PDF output languages:** Default setting for outputting languages in the PDF document to be generated
- **Revisions (property comparison of projects):** Settings for placing revision markers at connections
- **Revision (graphical representation):** Setting for the revision control of projects: What color or line thickness should be assigned to revision markers for changed, deleted or added objects or changes in the reports?
- **Revisions (change tracking):** Settings for revision control with change tracking, for example: What watermark is to be displayed on an incomplete page? Should a project be stored as a PDF upon completion?

- **Pages:** Defines the handling of pages. What combinations of letters and page names are allowed for subpages? How should paths be numbered: by page, by project or by structure identifier? The most important parameter in these settings is the definition or (subsequent) changing of the global plot frame for the project. A separate plot frame can be defined for every page in the page properties independently of the global plot frame setting. The page settings always have priority over the global project settings.
- **Symbol libraries:** Defines the symbol libraries used in this project. New entries (addition) of symbol libraries are automatically stored in the project after the setting is saved.

2.3.1.9 Setting Projects [project name]/Translation

This setting defines the databases to be used for translation or the scope of translation that EPLAN should use.

- **General:** This is where you enter project language settings, settings defining the handling of texts when translating during data entry. Settings for displaying translation languages (displaying translations in the project) are also configured here. You also have to define a source language here.

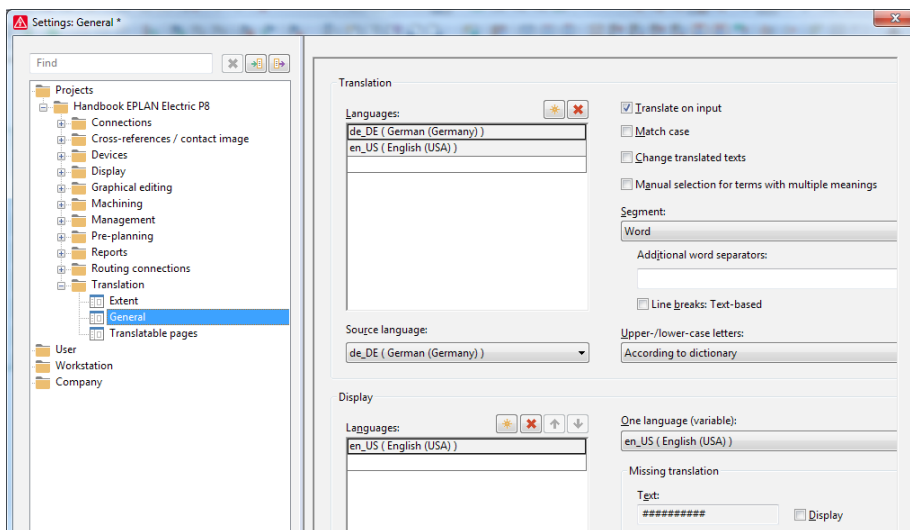


Fig. 2.25 Translation

- The definition of a **source language** is crucial, because this is the **first** language EPLAN displays in the various dialog boxes. This helps to prevent operating errors, e.g. entering English in the wrong “language line” by mistake.



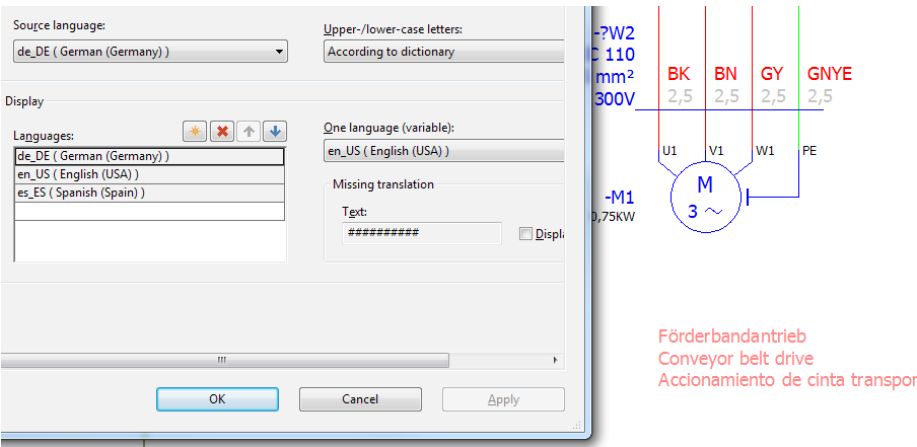


Fig. 2.26
Setting the translation sequence to “English, German, Spanish”

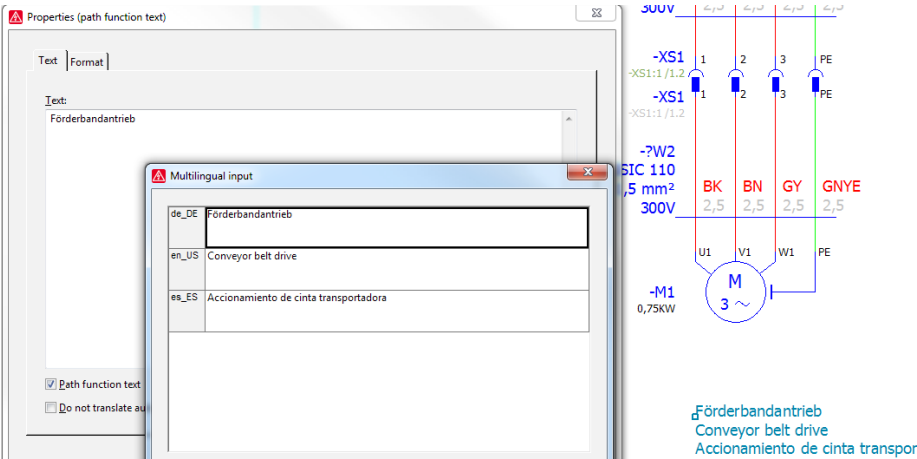


Fig. 2.27
Display in graphical editing

If the sequence of the *displayed languages* is changed in the settings, the *source language* will still be shown first in the dialogs.

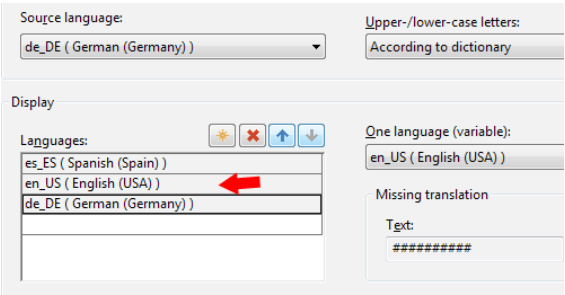


Fig. 2.28
Changing the translation sequence to “Spanish, English, German”

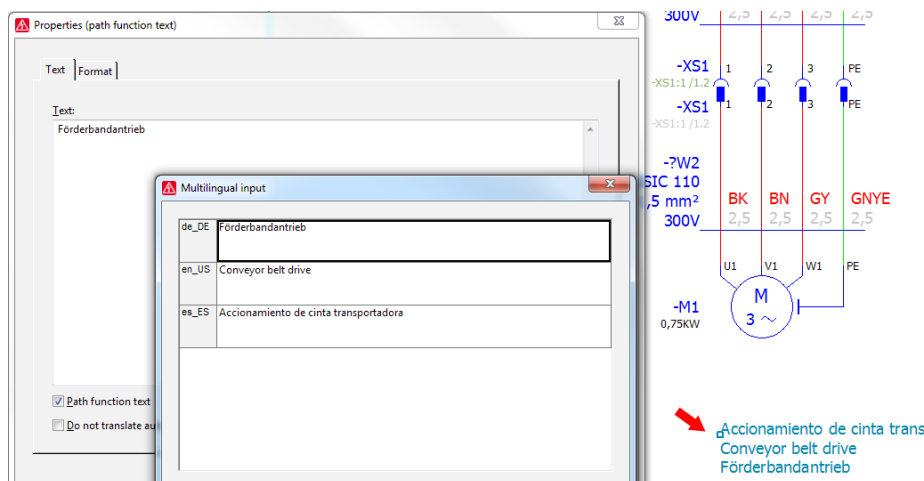


Fig. 2.29 Display in graphical editing

The sequence of the displayed languages has been changed correctly, but the sequence of the languages in the dialogs remains as is. This way, translations can no longer be entered by mistake.

Extent: This item allows the user to define the range of the properties to be translated for particular texts in forms, projects or components. Properties set to active are translated, and when the check box is deselected, they are not translated. This allows specific properties to be excluded from translation. The properties of the areas are defined by EPLAN and cannot be extended.

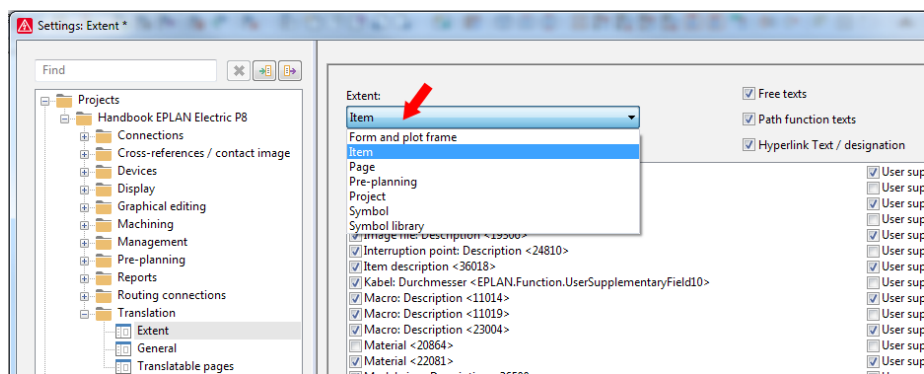


Fig. 2.30 Selection of settings for translation range

- **Translatable pages:** This setting is similar to the **Extent** setting. However, it generally relates to a page type. If page types are excluded from a translation, then they are not translated during an automatic translation run.

2.3.2 Settings – User

User settings relate to personal settings such as the user interface structure, which dialogs are opened on which screen, the directory structure for certain personal data, and the different representations of settings. All these settings depend on the login name entered at the Windows login and are stored in the directory `\\Basic\...\USER...CFG` in a workstation-specific manner (by computer). If you place these settings on a network drive, for example, then you can call up your own EPLAN configuration from any computer in the network.

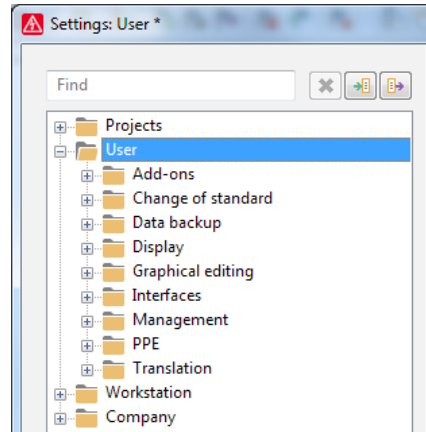


Fig. 2.31
User settings



The *User – Add-Ons* node contains settings for extension modules to be purchased separately. Since EPLAN has a fairly large selection of extension modules, which may vary for each license, this will not be discussed further at this point.

2.3.2.1 Setting User – Display

This user settings area contains the personal user interface settings or default values for a user-related workspace.

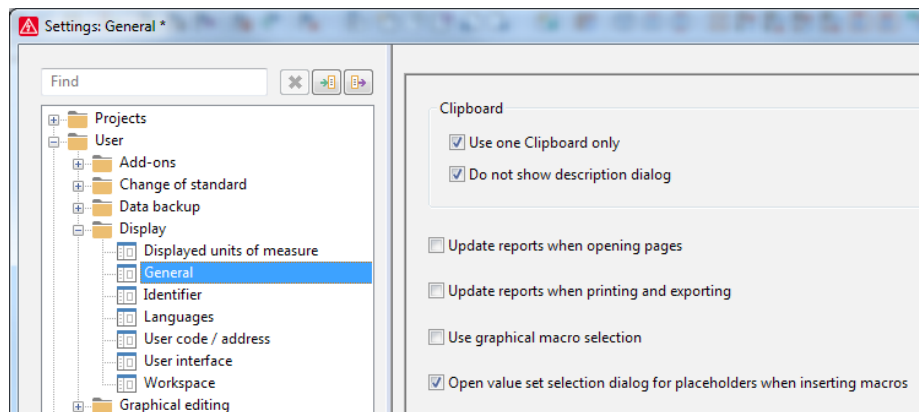


Fig. 2.32 General display settings

A number of important settings are explained below.

- **General:** Settings defining whether multiple clipboards or associated description dialogs should be used.

This settings area contains further settings, some of which are explained briefly below.

- **Update reports when opening pages:** If this setting is active, then reports that are already generated in the project and which exist as graphical output are automatically updated by EPLAN when a page of this type is opened.

This may be intentional, but it may also be the case that these reports should not be graphically changed by EPLAN, e.g. for revision reasons or for tracking changes. This parameter should then be switched off by deselecting the option check box.

This parameter should also be switched off for performance reasons if project editing becomes very slow (EPLAN must constantly update the data in the background and then update the report as soon as it is opened – this requires a fast computer).

This also applies to the **Update connections when switching pages** setting. The **Update connections throughout the project (manually)** setting has the following effect: If this setting is active, the connections will be updated throughout the entire project. If the setting is inactive, only the selected connections (one or several pages or selected devices, etc.) will be updated.

- **Update reports when printing/exporting settings:** This way, the reports are always up-to-date when the project is printed or exported.
- **Use graphical macro selection:** This setting can only be used in combination with the *Company/Display/Graphical macro selection* node settings.

The node also contains these settings

- **Displayed units of measure:** Here you define the units of length and weight as well as the number of displayed decimal places.
- **Workspace:** This is where you define the preferred user workspace. The workspace can be changed at any time during project editing.
Workspaces will be handled in more detail later in section.
- **User code/Address:** This is where you enter the user's code, name, phone and e-mail address. Some of these entries are used for the page or project properties, and others are used for system messages, for example when a conflict with another user occurs during project editing. These entries are also useful for forms, plot frames, etc.
- **Identifiers:** Default settings defining how EPLAN should sort structure identifiers such as higher-level function or installation site that are reassigned during project editing

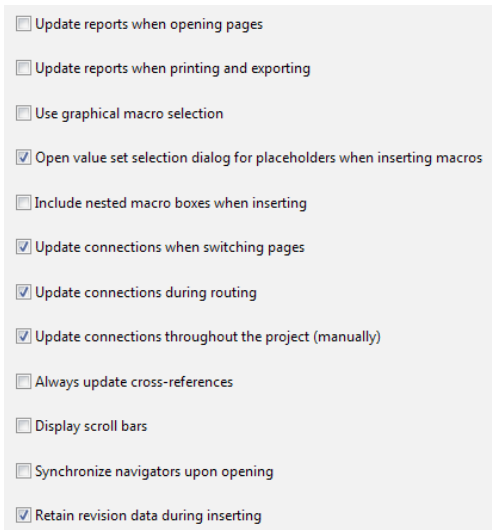
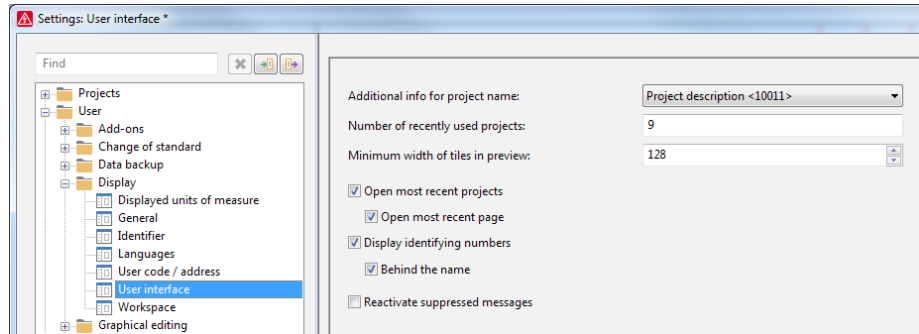


Fig. 2.33
Additional general settings

Fig. 2.34
Extended interface
settings



- **User interface:** General settings, such as regarding project information and/or the display of preview information.

Further user interface settings are possible here: Should the last projects be reopened when the program starts, or should the property number (identifying numbers) be displayed with the properties in EPLAN (= my recommendation)?

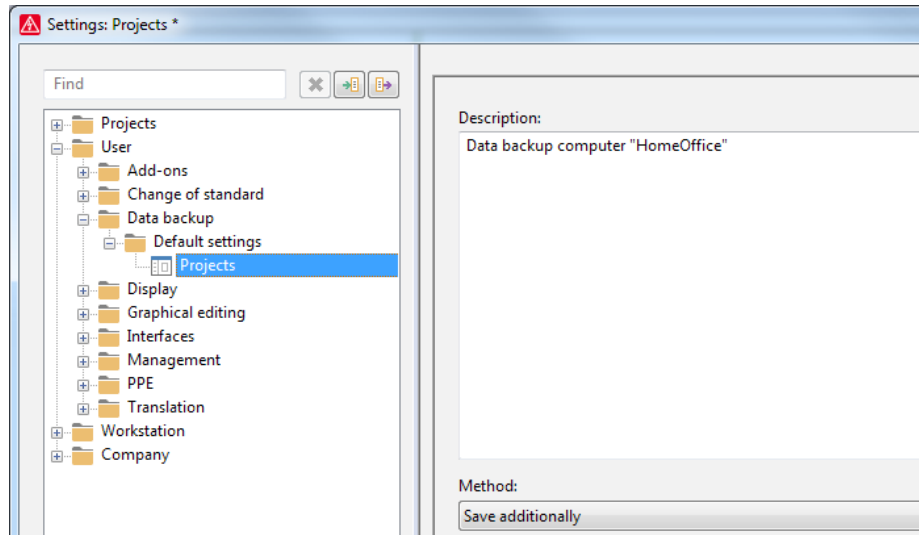
The *Reactivate suppressed messages* setting is interesting. If this is set, then dialogs whose displays have been deactivated are once more displayed.

- **Languages:** Language setting of the EPLAN system and online help system setting. This is used, for example, to define whether the online help system is to be used locally or online (requires Internet connection).

2.3.2.2 Setting User – Data backup

These user settings are for defining default values for backing up master data.

Fig. 2.35
Data backup settings



- **Default settings/Projects:** Default settings for project backup details such as the e-mail message split size or the backup drive.

2.3.2.3 Setting User – Graphical editing

These settings define further user interface settings such as colors or settings for placing connection symbols (T-nodes, etc.).

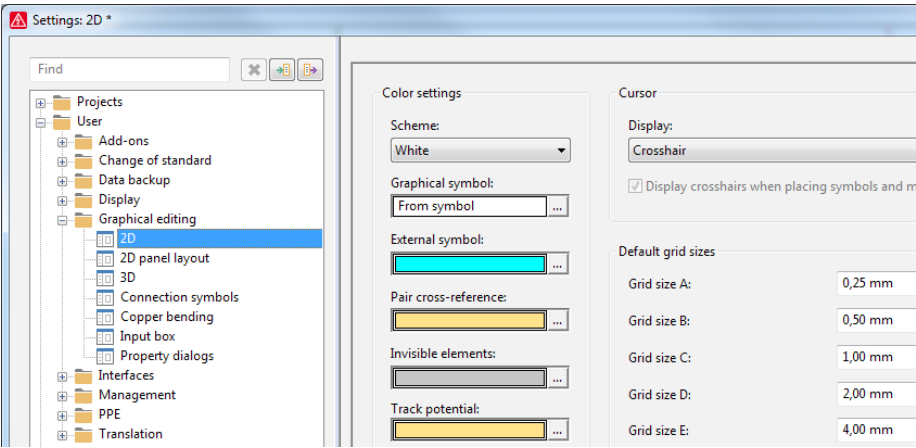


Fig. 2.36
Settings for graphical editing

- **2D:** Display of color settings, background color, grid sizes, cursor settings, and scrolling behavior

The following settings, especially the grid sizes, have proven useful in practice. The 4; 2; 1; 0.5 and 0.25 levels have proved particularly practical, especially when separate keyboard shortcuts are used for the grid settings.

- **2D panel layout:** Settings for part placements, their mounting options, and settings for applying dimensions from the parts master data or manual input
- **3D:** Settings for the 3D area, such as grids, color settings and how terminal strips are to be handled during placement
- **Property dialogs:** Settings defining whether certain property dialogs, for example **Shield properties**, should be automatically opened after an object is placed
- **Connectors:** Important settings for defining whether or not the connection dialog should be displayed before a connection junction (T-node) is placed and whether or not changes to connection designations should be transferred to subsequent connections

2.3.2.4 Setting User – Change of standard

This is an important area for projects impacted by a change of standard. There will not be any entries in this node either until a change of standard has occurred.

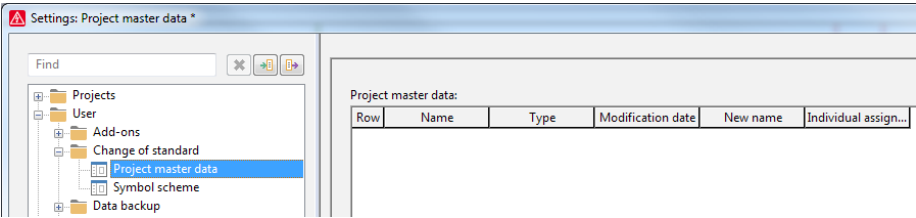


Fig. 2.37
Change of standard

- **Project master data:** Contains project master data assignments. This area is only filled with data when a change of standard has occurred.
- **Symbol scheme:** Contains symbol assignments. Similar to project master data, if no change of standard has been performed then this area is empty.

2.3.2.5 Setting User – Interfaces

EPLAN offers a number of different interfaces for importing and exporting data. The interface settings are defined in this area.

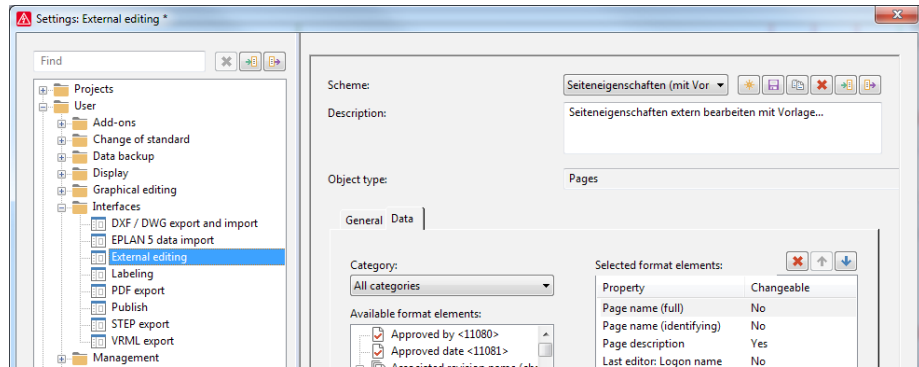


Fig. 2.38 Various interface settings

These settings are adopted as default values if no other settings are used during the direct output.

- **Labeling:** This is where you define default settings for the most frequently used schemes, in order to transfer data to the labeling system.
- **Export and import of DXF/DWG:** Default settings for importing and exporting DXF/DWG files
- **EPLAN 5 data import:** Settings for importing EPLAN 5.x data.
- **Edit externally:** Definition of various schemes for external editing. These settings affect a number of different areas, such as functions, pages, and connections.
- **PDF export:** Settings for internal PDF output of pages or projects.
- **Publish:** Settings for EPLAN's own publication EPDZ format. This allows for EPLAN projects to be viewed using the EPLAN View app.

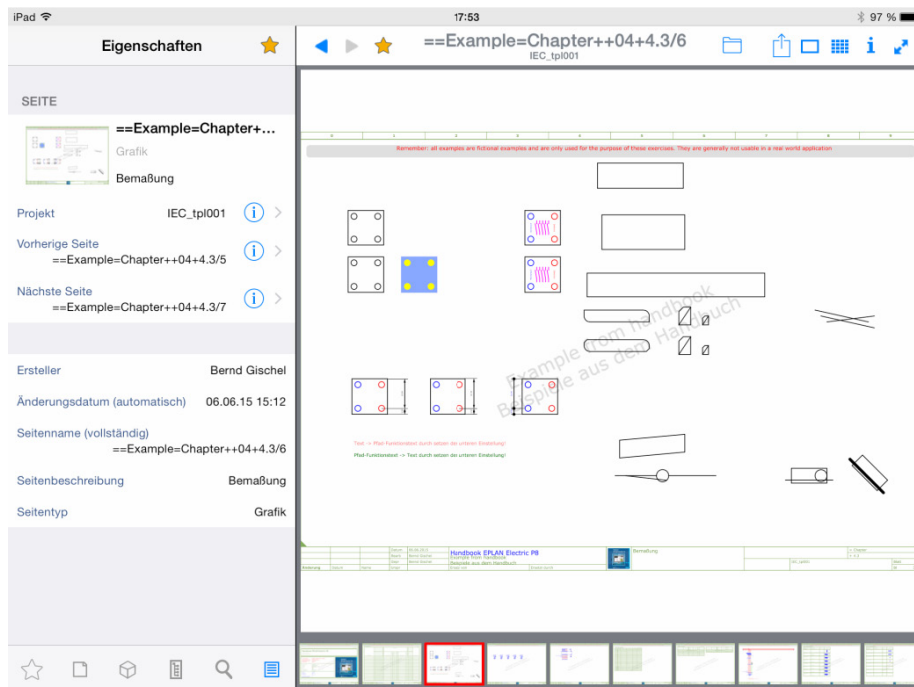


Fig. 2.39 Example: EPLAN project published in the EPLAN View app (iPad)

2.3.2.6 Setting User – Management

In this area, EPLAN allows user-specific settings for management tasks.

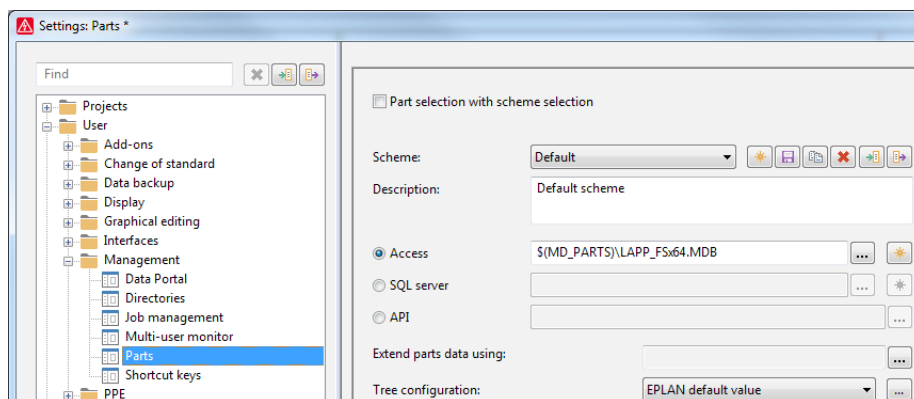


Fig. 2.40 Management settings

- **Part:** Defines part selection, where the parts are selected and the interface to be used (API, internal, SQL server or ODBC interface). This can also be an external PPC system.

- **Data portal:** Settings for the user, the portal and the connection settings to the portal.
- **Shortcut keys:** User-specific settings for defining shortcut keys for EPLAN commands. The shortcut keys can be individually assigned or deleted here, or the keyboard assignments can be restored to the default settings.
- **Directories:** A complete overview of the installed directories for the various master data and databases in EPLAN. You can also use your own schemes to change all the directories in a single step.

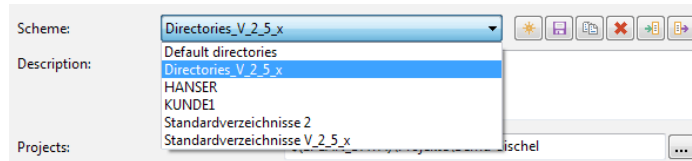


Fig. 2.41 Default directories

2.3.2.7 Setting User – Translation

These are general settings for defining how languages and translations are to be handled.

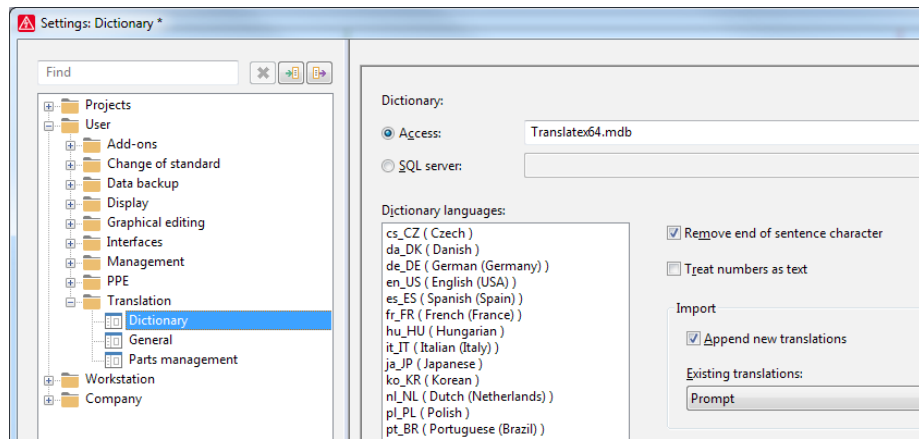


Fig. 2.42 Translations and foreign languages

- **General:** Definition of how project-independent texts are to be translated and which source language is to be used. The source language is the language for which translations will be later generated.
- **Parts management:** Definition of how describing part texts are to be translated in the translation run of parts management.
- **Dictionary:** Defines the dictionary the user would like to work with. The foreign languages available in the database are displayed simultaneously. The parameter also allows default settings for importing new translations and defines the behavior when texts are entered (AutoComplete or AutoCorrect).

2.3.3 Settings – Station

These are general settings for the current workstation.

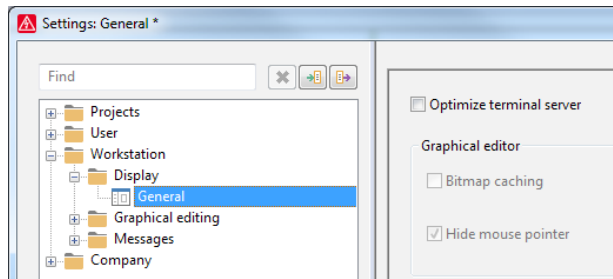


Fig. 2.43 Station settings

- **Display/General:** This allows you to set operational optimization if you are using a terminal server.
- **Graphical editing/Print – PDF output size:** Here you can set various global options related to printing, such as print margins or print size, and enter settings that affect the size (scaling) of a PDF output.
- **Messages/System:** This is where you define the path to, and the maximum size of, the system messages file.

2.3.4 Settings – Company

The company settings contain a number of parameters that, for example, can make comparisons of projects from external suppliers easier.

- **Display/Graphical macro selection:** This setting allows you to more easily find and select macros and their directories using your own texts (descriptions) and/or images.

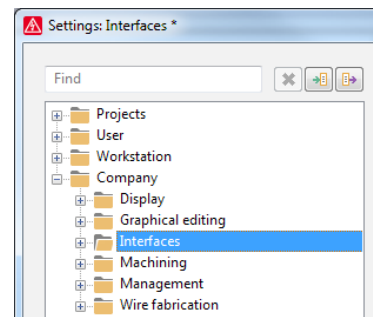
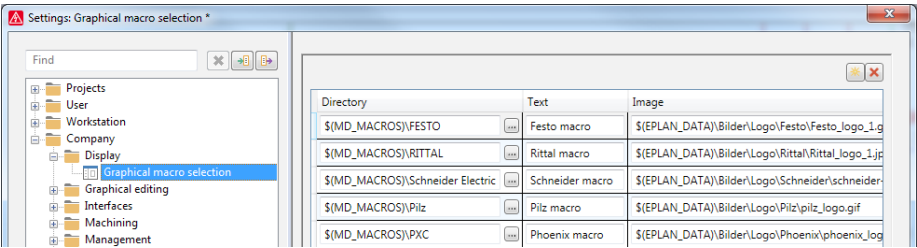


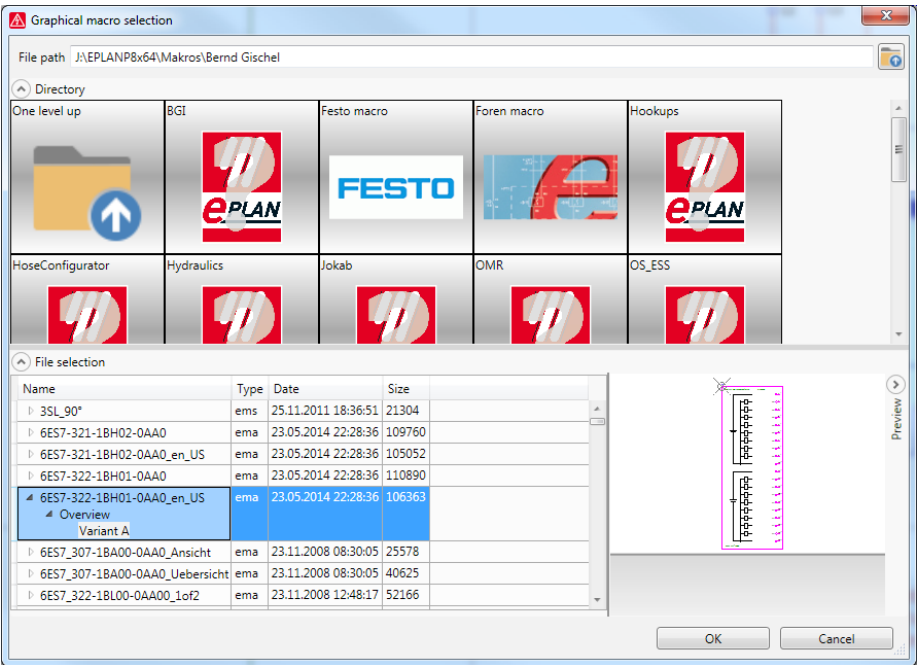
Fig. 2.44
Company settings

Fig. 2.45
Defining the settings
for graphical macro
selection



- These default settings can only be used in the *graphical editor* if the **Use graphical macro selection** option has been selected in the *User/Display/General* settings.

Fig. 2.46
Graphical macro
selection dialog
(started from the
graphical editor)



- **Graphical editing/Fonts:** Defaults for up to ten company fonts. These ten different fonts allow you to later switch from one font to another in the entire project without any problems, provided the affected object uses one of the global fonts. However, priority is always given to the project-specific fonts (setting under *Projects/[project name]/Graphical editing/Fonts*).
- **Management/Property comparison of projects):** Settings determining the properties to be used for project comparison. You can create your own schemes for comparisons.
- **Management/Project management database:** This is where you can set the storage location of the company-specific project database.

- **Management/Add-ons:** Allows you to automatically control EPLAN Electric P8 when it is started using the manual defaults defined in an *Install.xml* file, whose location is defined here.
- **Management/Change tracking (numbering of the revision index):** Settings (controllable via your own schemes) for indexing revisions

■ 2.4 EPLAN and multiple starts?

EPLAN can be started as often as desired. You may ask why, since you can work on several projects at the same time in EPLAN. In general, you do not need to start EPLAN multiple times, but it is possible.

One of the practical benefits of EPLAN multiple starts is that you can have parts management and the dictionary open at the same time, allowing you to enter missing entries without having to interrupt your work on open projects.

When you are finished editing, for example in parts management, you of course should and must synchronize this data with the project and decide where these parts are to be stored. This can be done automatically on each subsequent start. Here too, you must carefully consider whether automation at this point is actually useful. EPLAN lets the user decide.

■ 2.5 Properties

In addition to the actual graphics of symbols, forms, or plot frames, EPLAN also outputs logical information. This logical information must be specified for the symbols, forms, or plot frames so that it is visible. This is done by assigning properties.

Every property has a property name and an associated property number (known as an ID, and possibly also an index) that is only valid for this specific property. You can use the property number to gain an *approximate* idea of the area that the property belongs to (project, page, etc.). The property value assigned to each property is unique to the property.

The different types of properties in EPLAN

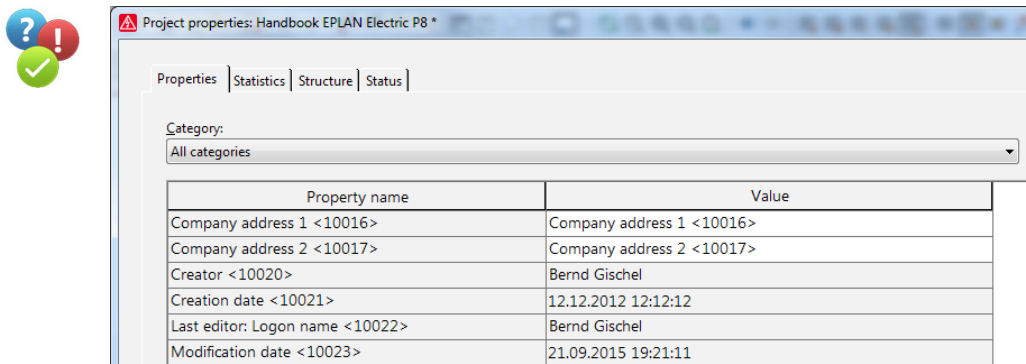


Fig. 2.47 Project properties

The *Project name* <10000> property contains the <Project name> value (the name of the project).

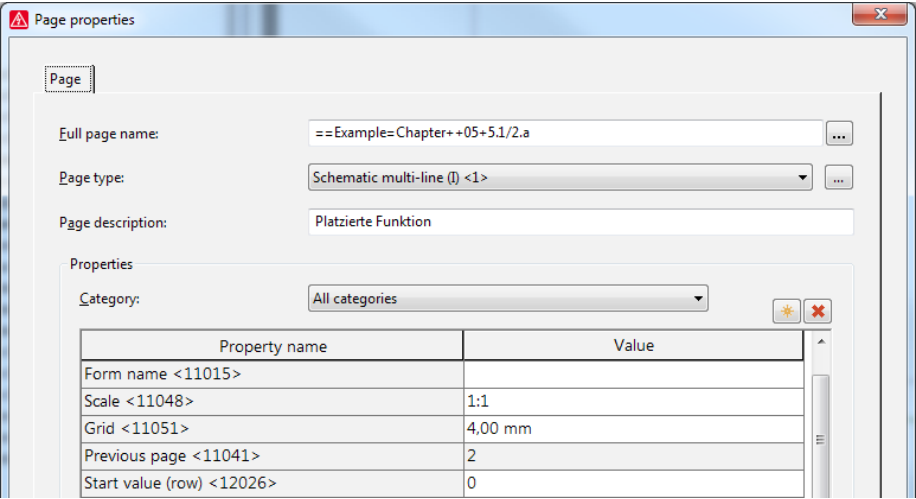


Fig. 2.48 Page properties

The *Page number* <11042> property contains the <n> value (n = the actual page number). EPLAN generally distinguishes between several different types of properties. There are *Project properties* (relevant project information such as the project name; 10000), *Page properties* (properties of a page in the project itself, e.g. the current page number; 11042), *Symbol properties* (property of a symbol such as the displayed device tag; 20010), *Form properties* (properties for constructing a form and reporting data such as terminal and pin designations; 20030), *Plot frame properties* (e.g. the search direction for transferring the device tag, whether the device tag should be automatically transferred from the first device to the left; 12103), and numerous other function properties.

As an addition to the above fixed, not extendable, properties, EPLAN has integrated the user-defined properties that can be edited and configured for your own requirements.

2.5.1 Project properties

Project properties can also be described as global properties. These properties can be used everywhere as special text (project property), in graphical editing for example via the **INSERT/SPECIAL TEXT/PROJECT PROPERTIES** menu item. The **Special text – Project properties** dialog will open.

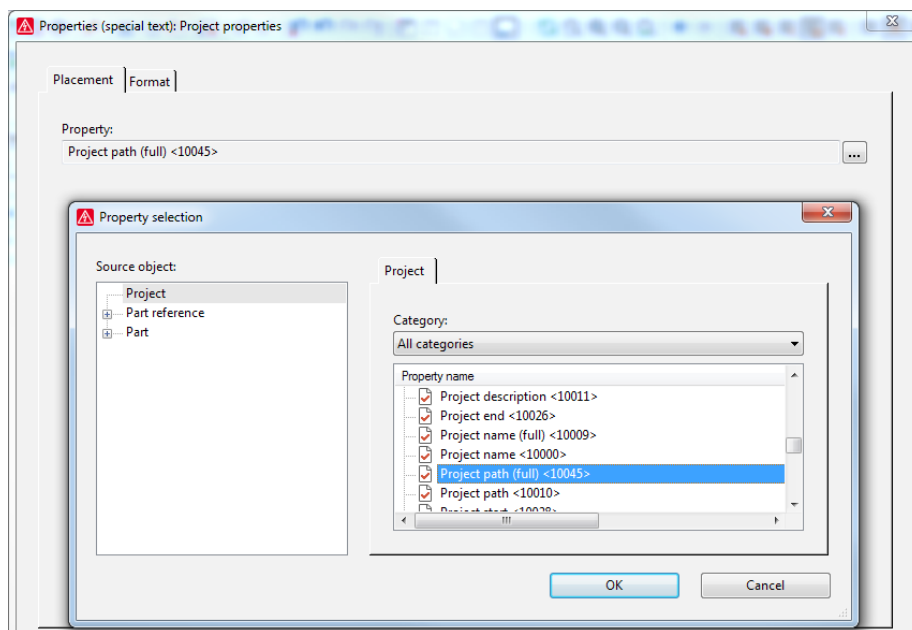


Fig. 2.49 Selecting and inserting a project property

Clicking the  button opens the **Property selection** dialog. Here you can select the desired project property and accept it by clicking OK. The project property is loaded into the *Property* field of the **Special text – Project properties** dialog.

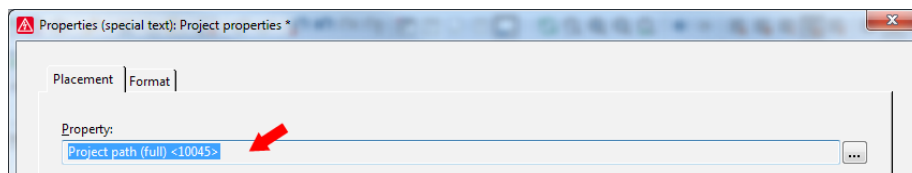


Fig. 2.50 Transferred project property

The special text can then be formatted (font, width, italics, etc.) on the **Format** tab. When you have finished entering data, you confirm by clicking **OK**. The project property now hangs on the cursor and can be placed.

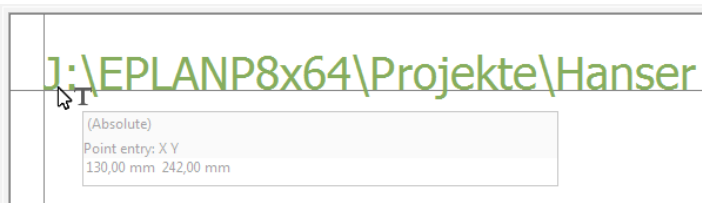


Fig. 2.51 Placing project properties

Project properties can be added to the project via the  graphical button in the project properties. Use the **PROJECT/PROPERTIES** menu to open the project properties. Now click the  button to open the **Property selection** dialog.

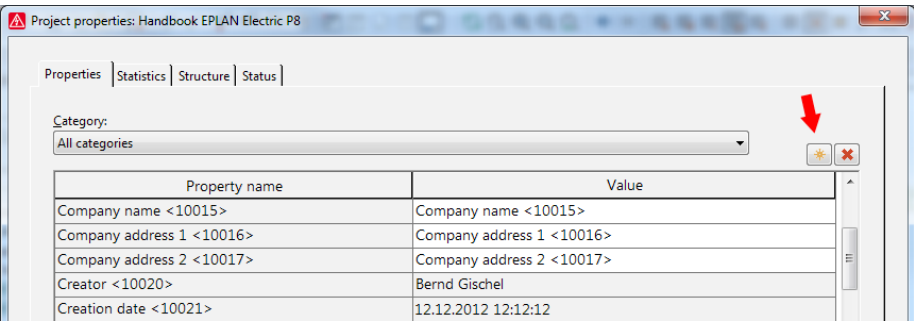


Fig. 2.52 Add project property

In the **Property selection** dialog, the usual Windows functions can be used to select one or more properties and these are then loaded into the project properties of the project by clicking the **OK** button.

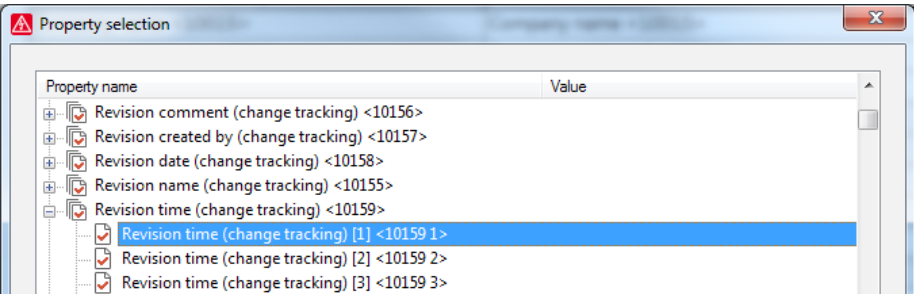


Fig. 2.53 Property selection

You can use the right-click popup menu to sort the display of the project properties. Select the **CONFIGURE** command in the popup menu. The **Property arrangement** dialog opens. The graphical buttons (blue arrows up or down) can be used to arrange the properties as you wish.

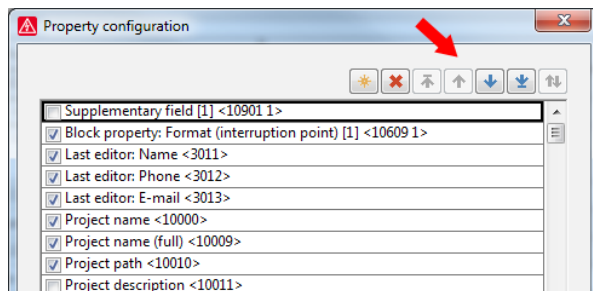


Fig. 2.54 Property configuration

2.5.2 Page properties

EPLAN **page properties** are those properties belonging to a page. Page properties can also be used everywhere. The **INSERT/SPECIAL TEXTS/PAGE PROPERTIES** menu item in graphical editing opens the **Special text — Page properties** dialog. In the *Property* field, you can use the  button to open the **Property selection** dialog.

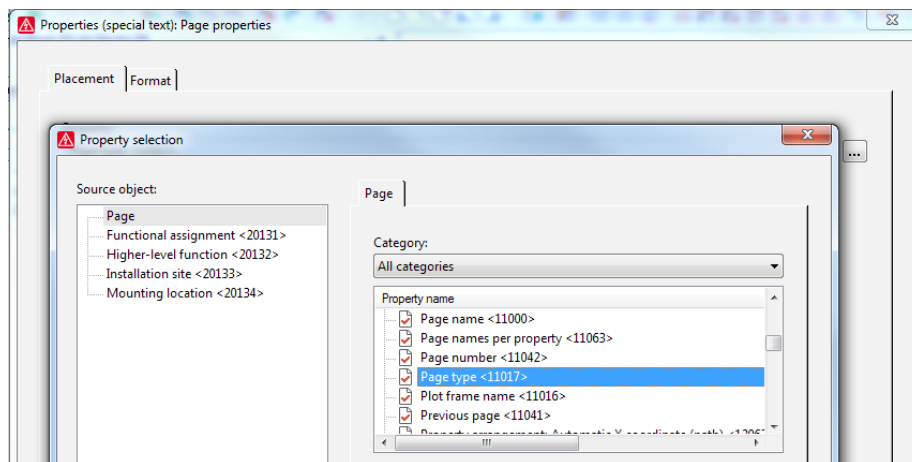


Fig. 2.55 Add page properties

Then you select the desired property. When you click **OK**, the selected property is loaded into the *Property* field of the **Special text — Page properties** dialog.

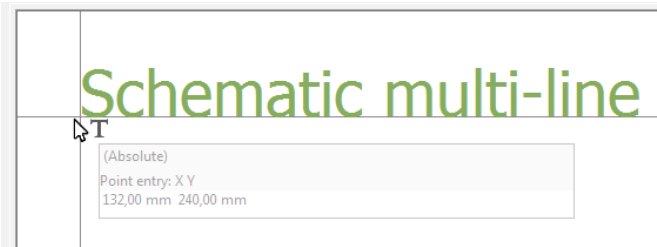


Fig. 2.56
Placing a page
property

After formatting the text (if necessary, on the *Format* tab), you close the **Special text – Page properties** by clicking OK. The special text now hangs on the cursor and can be placed as desired.

2.5.3 User-defined properties

User-defined properties are properties that the user can create and edit freely. User-defined properties are always created in a project-specific manner, but can be easily integrated with or added to other projects by means of EPLAN's integrated import and export functions. User-defined properties are assigned to specific areas (project, page, etc.). Using default selection lists, self-extending selection lists or free entries, they can be created and subsequently selected easily.

User-defined properties are edited using the **CONFIGURE OPTIONS/PROPERTIES** menu.

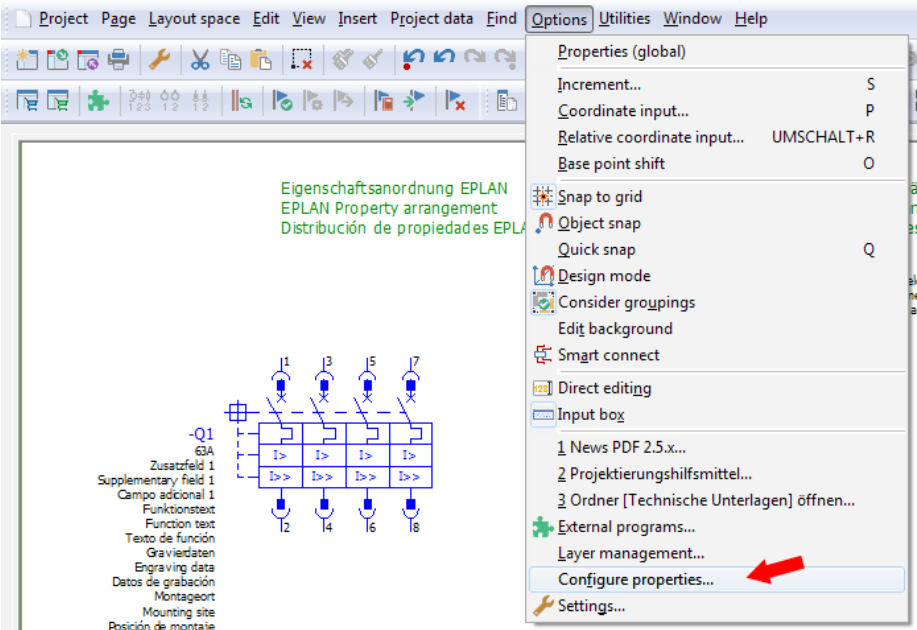


Fig. 2.57
Menu call: configuring
(user-defined)
properties

Afterwards, EPLAN opens the **Configure properties** dialog. If there are no user-defined properties yet, the left area of the dialog will remain empty.

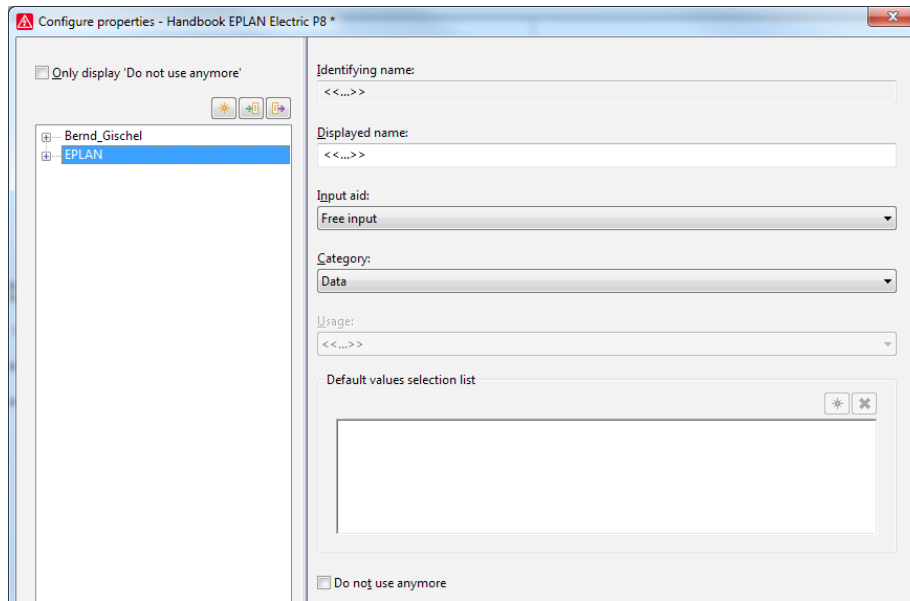


Fig. 2.58 Dialog for configuring properties

Click the **NEW** button (☀ sun symbol) in the left area to create a new user-defined property and configure it for subsequent use. EPLAN opens the **Generate user-defined property** dialog.

Now, the new property can be specified. Depending on the input, EPLAN differentiates several levels. Each new level must be separated by a period.

In the example, the identifying name is thus *Level1*, followed by *Level2*, then *Level3*, etc.

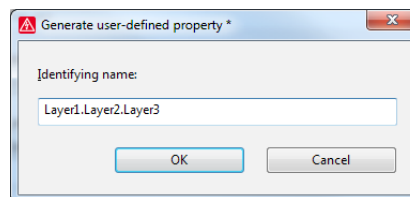


Fig. 2.59
Dialog Generate
user-defined property

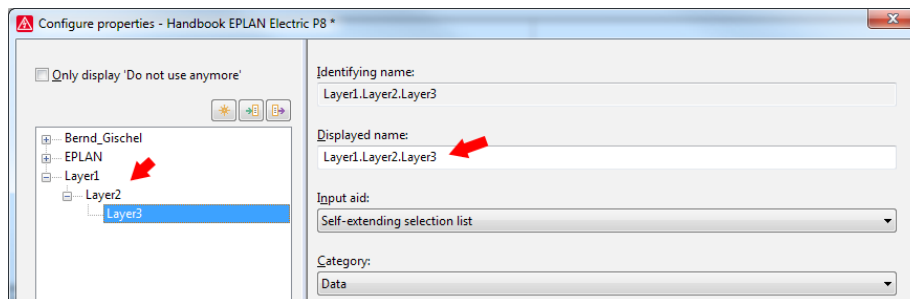


Fig. 2.60 Structure of the identifying name

This is how a structure can be established. Similar user-defined properties are sorted into the structure correctly.

The user-defined property can now be configured further. The following settings are possible:

- **Displayed name:** This can be edited freely and allows for multiple languages
- **Input aid:** Selection of one of the three options (free entry, self-extending selection list, selection list)
- **Category:** Selection of a category (such as devices, data, etc.) to be assigned to the user-defined property
- **Usage:** Selection of objects (project, page, etc.) to which the user-defined property is to be apply

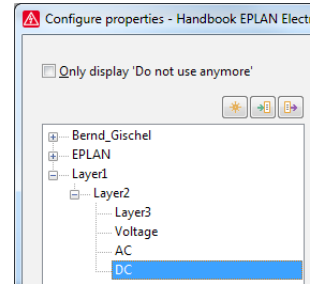


Fig. 2.61
Structure



NOTE: Once the user-defined property has been saved, the Usage selection can no longer be changed.

For example, a selection list (5V DC, 12V DC, etc.) with values can be specified for a user-defined property (control voltage DC) for the Data category and Project (Usage) object to limit the freedom in creating the project.

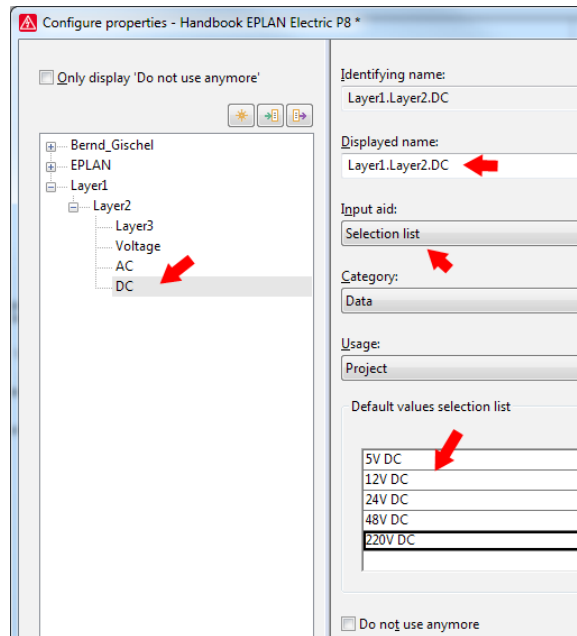


Fig. 2.62 Default values from a selection list

The user-defined property is added via the **NEW** button, as is common EPLAN practice.

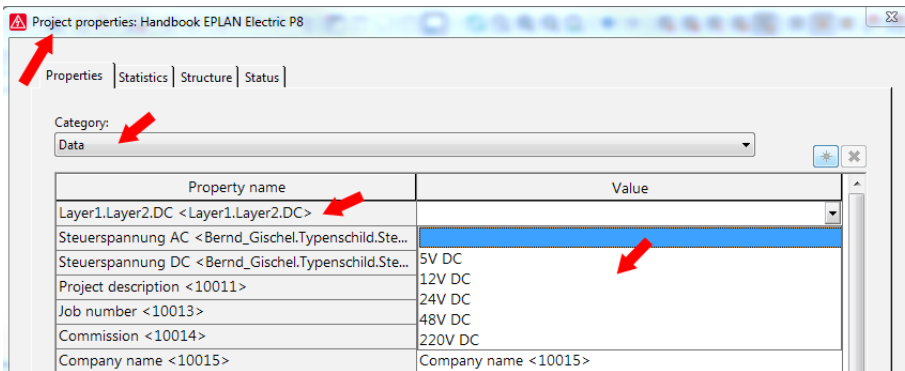


Fig. 2.63 Selection default value

2.5.3.1 Remove user-defined properties

To remove a user-defined property from a project: in the **Configure properties** dialog (CONFIGURE OPTIONS/PROPERTIES menu), select the *Do not use anymore* checkbox for this property.

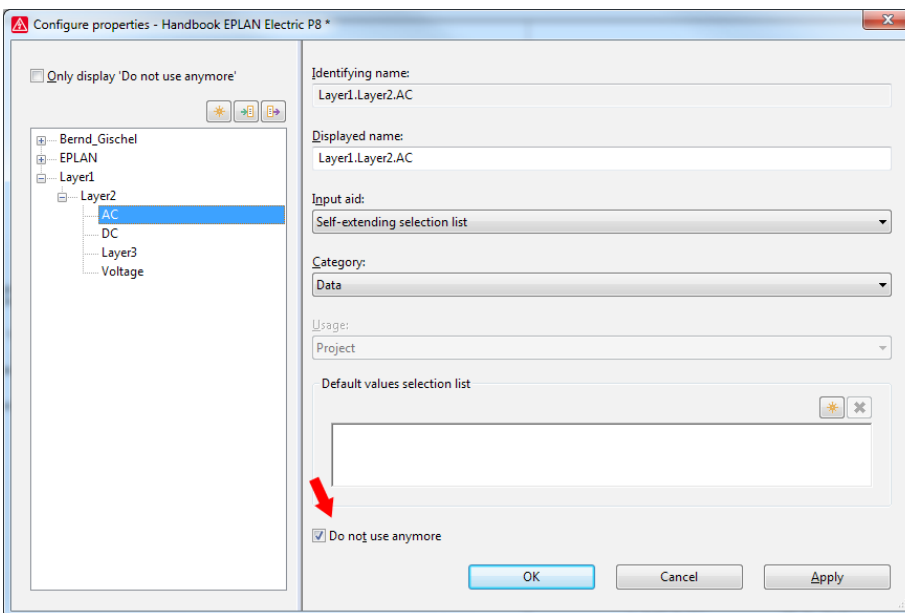


Fig. 2.64 Checkbox Do not use anymore

Entire nodes can be selected as well before selecting the *Do not use anymore* checkbox. These selected user-defined properties can be removed from the project definitively with one compression run.

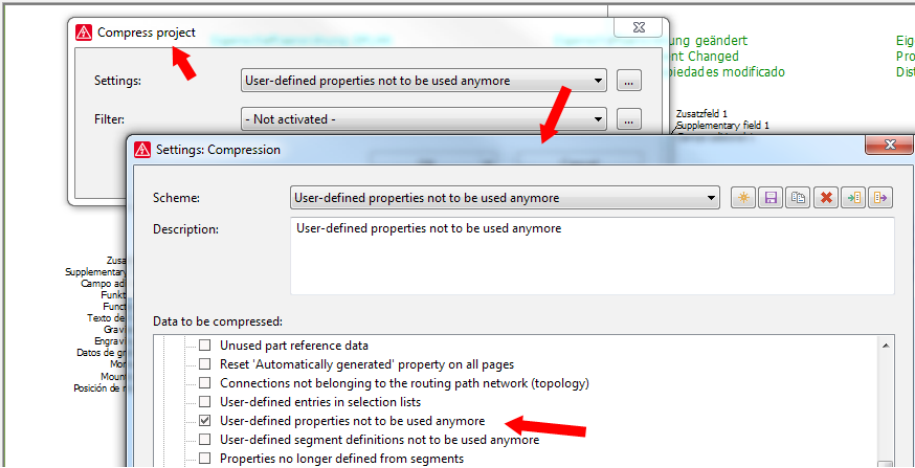


Fig. 2.65 Compression run

2.5.4 Symbol properties (components)

Symbol properties are properties that are assigned to special symbols (components). Some symbol properties, such as connection point designations or device tags, can be accessed directly.

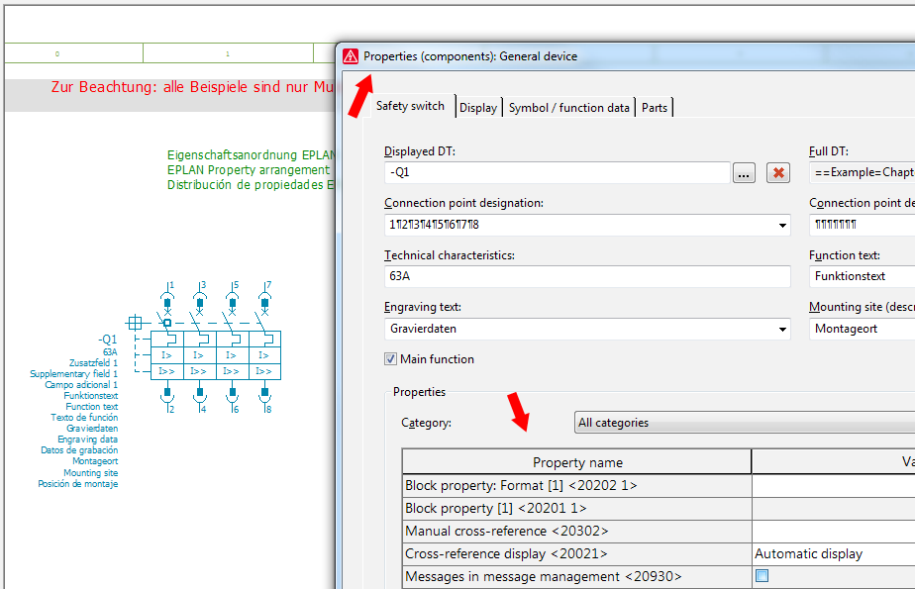


Fig. 2.66 Properties of a symbol (component)

New properties are added in the same way as other properties (project properties or page properties). You click the  graphical button on the *Property (device type)* tab. EPLAN opens the **Property selection** dialog. Here you can select the desired property or properties to be used in the symbol properties and confirm by pressing **OK**. The property has now been added.

It is also possible to adjust the sequence in which properties are displayed in a symbol. You call up the popup menu using the right mouse button. Select the **CONFIGURE** entry in the menu that opens up. EPLAN opens the **Property arrangement** dialog. Here you can now use the familiar graphical buttons to move or sort the properties.

2.5.5 Form properties

In EPLAN, **Form properties** are the properties of a form. Forms (reports) are distinguished by report type. Some properties are shared by all types of forms, but there are also properties that apply only to specific report types. Form properties can be edited only in the **Form editor**.

2.5.6 Plot frame properties

Plot frame properties are properties assigned to the structure and reports of normal page types, such as the *schematic multi-line*. An example of this is the *Path areas* property and its size. Plot frame properties can be edited only in the **Plot frame editor**.

■ 2.6 Buttons and popup menus

EPLAN has many dialogs. This is not unusual in such an extensive program. At this point, I would like to draw special attention to a number of dialog elements that occur repeatedly in many dialogs. These dialogs are similar in many different places in EPLAN. It is therefore important to know what they mean and how they can be used.

Graphical buttons play an important role because they provide easy access to most of the functions and procedures in EPLAN.

2.6.1 Device dialog buttons

Apart from a few small details, all EPLAN device dialogs are very similar. This means that they always have the same structure, regardless of whether it is a device dialog for a motor overload switch or a transformer or any other component. EPLAN uses different buttons in different tabs to simplify operation. Here is a list of the most commonly used buttons and what they mean.

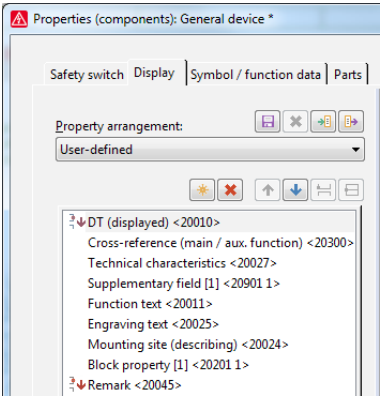


Fig. 2.67
Various buttons

-  Add new property to property arrangement
-  Remove property from property arrangement
-  Move property up one place
-  Move property down one place
-  Undock property (release)
-  Dock property (link to upper area)
-  Save
-  Delete
-  Import
-  Export

2.6.2 Buttons in dialogs (configuring)

You can use the **CONFIGURE** menu item to better arrange or re-sort properties in dialogs to suit your personal needs.

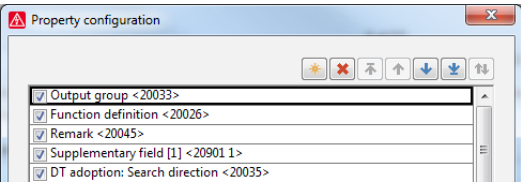


Fig. 2.68
Buttons in the property
configuration dialog

These can be page properties and also settings in other dialogs such as the **UTILITIES/GENERATE REPORTS/SETTINGS/OUTPUT TO PAGES** dialog. The number of buttons can vary here. In principle, the meaning/function of the buttons remains the same.

-  Move the selected entries to the start of the list
-  Move the selected entries to the end of the list
-  Swap two selected entries (swap positions)

In contrast to the graphical buttons in the device dialog, multiple entries can be selected here. The functions provided by the other graphical buttons have already been described.

2.6.3 Buttons in dialogs such as filter schemes

In addition to the already familiar buttons, dialogs such as **Filter**, **Sorting** and other settings dialogs have additional graphical buttons.

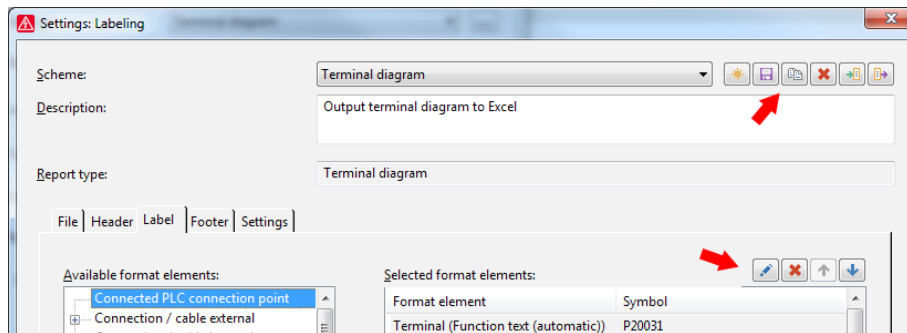


Fig. 2.69 Buttons in various settings dialogs

Additional buttons in the **Filter** or **Sorting** dialogs:

-  Edit entry
-  Copy entry

2.6.4 Restore default function

Depending on the dialog, you may find the *Restore default* function on the right mouse button. To call this function, click the corresponding field (such as Path selection) and then right-click. It is very useful in dialogs that open files or directories. Here you can quickly set the default directories, without having to click all the way through to the default directory.

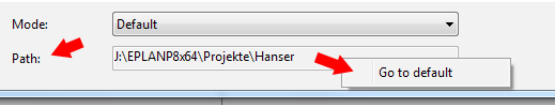


Fig. 2.70 Restore default function

2.6.5 Property arrangements (components)

EPLAN always offers a *Default setting* for property arrangements. The default setting is defined in the symbol structure and cannot be changed here. To change these default settings, you have to edit the symbols. I recommend that you do not make any changes to the symbol libraries that came with EPLAN since changes may make them incompatible with newer versions of EPLAN symbol libraries, making synchronization impossible. You can also change these property arrangements. Once they have been changed, they initially appear in the property arrangement with the designation *User-defined*.

To undo this user-defined property arrangement, simply select the *Default setting* from the *Property arrangement* selection list. This returns all manually changed settings on a symbol back to the default values defined for the symbol.

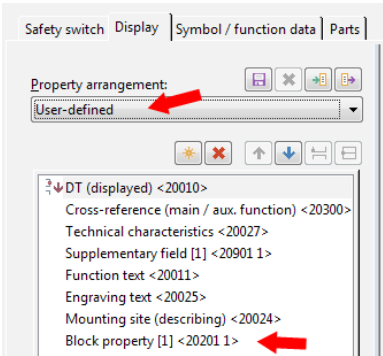
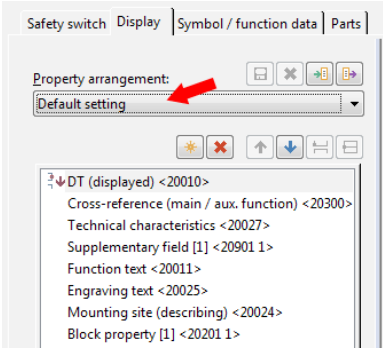


Fig. 2.71
Default setting

Fig. 2.72
User-defined
arrangement

If a property has been changed, then EPLAN no longer regards it as a default value and the display in the *Property arrangement* selection field changes to *User-defined*. This user-defined arrangement can also be saved via the **SAVE** button and then called up or set at similar symbols (functions). You can assign a descriptive name to the property arrangement. After you click **OK**, this property arrangement is saved, entered into the selection field and is then available to all symbol variants.

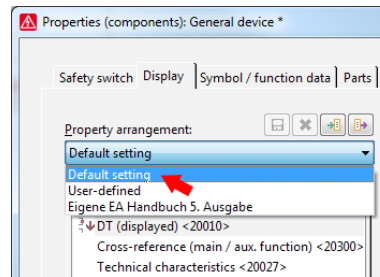


Fig. 2.73
Undo

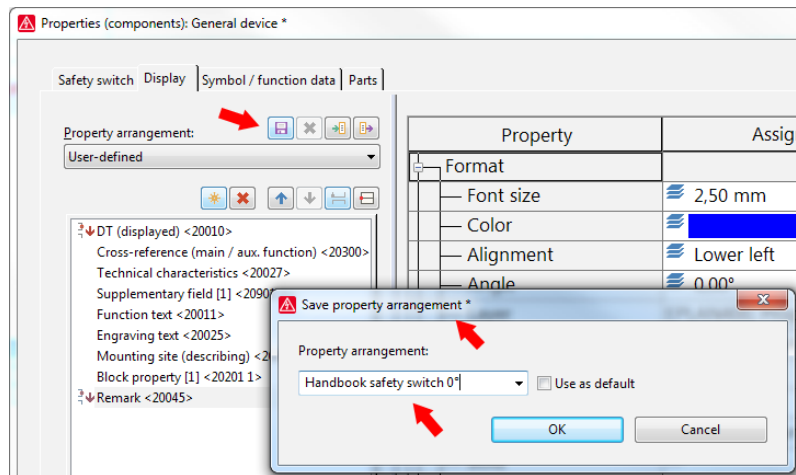


Fig. 2.74 Saving your own property arrangement

Property arrangements can only be transferred to the same “types” of devices (same function template), i.e. the property arrangement of a safety switch cannot be transferred to the property arrangement of a terminal.

The *Use as default* option allows EPLAN to immediately (the next time you insert the same symbol) and automatically use your own property arrangement.

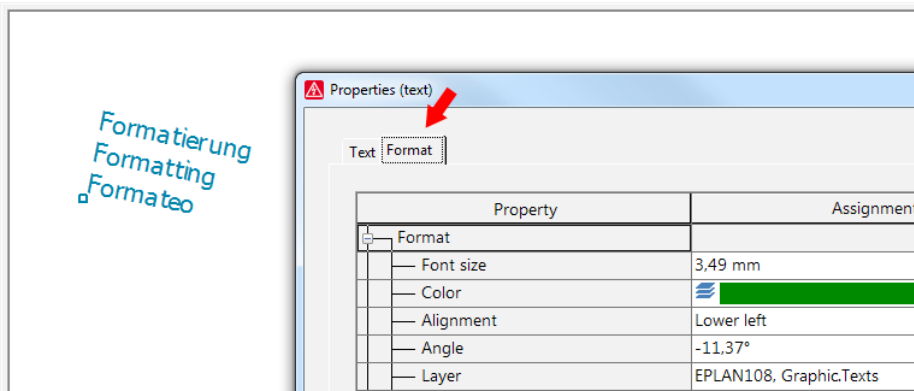
The property arrangements can also be imported or exported in order to continue using them in other projects. Use the **PROJECT/ORGANIZE/PROPERTY ARRANGEMENTS** menu for import/export.

2.6.6 Format properties

EPLAN makes it easy to use different formats, e.g. for texts. Though the dialogs may be somewhat different, the basic principle always remains very similar.

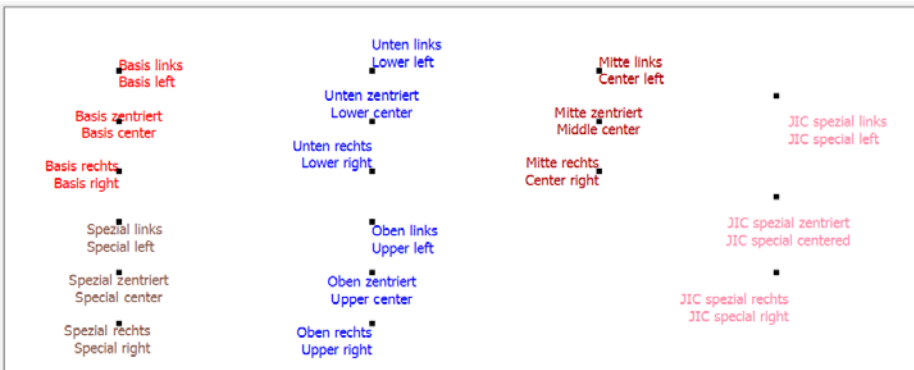
A normal text dialog is used as an example here. To insert a text, you use the **INSERT/GRAPHIC/TEXT** menu to open the **Properties** text dialog.

Fig. 2.75
Formatting options for texts (example)



You can now enter the desired text into the **Enter text** field on the *Text* tab in the opened dialog. To format the text according to your personal wishes, you need to switch to the **Format** tab. You can freely format the text you just entered. You can edit all selection fields or select default values.

Fig. 2.76
Examples of different formats



This example shows how different the arrangement and appearance of a text can be if it is formatted using the default setting *Alignment* (the small black dots are the anchor points of the text).

Additional representation in dialogs, including the option of changing formatting (font size, angle, etc.):



This symbol at the start of a line (in the *Value* column) means that the value entered is the original value from the corresponding level and has not been modified. If the value is changed, this symbol will disappear.

2.6.7 Buttons (small black triangles)

In a number of dialogs, the buttons have a small black triangle. Additional menus are “hidden” behind these buttons.

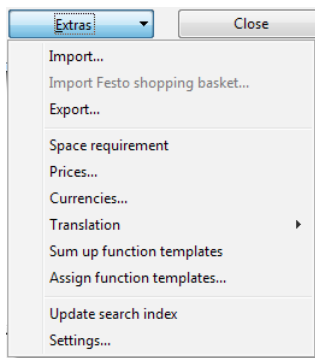


Fig. 2.77 Parts management

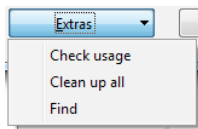


Fig. 2.78 Structure identifier management

2.6.8 Dialogs for schemes

Dialogs for schemes allow you to export or import created schemes.

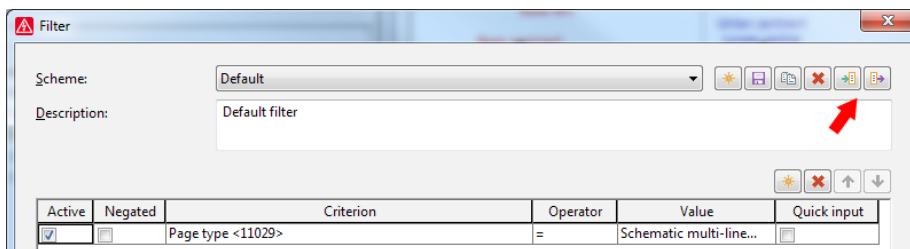


Fig. 2.79 Import and export options for schemes



TIP: User-created schemes should be regularly exported, preferably as soon as they have been created. These schemes are then always available for new versions, other projects or other workstations (depending on the installation).

In many areas, EPLAN offers a special identifier for the scheme type. This way, during an import, only those scheme types that can actually be imported into the current scheme are displayed.



<i>CBnu. *.xml</i>	Example: Scheme prefix for cable numbering
<i>LB. *.xml</i>	Example: Scheme prefix for labeling
<i>WS. *.xml</i>	Example: Scheme prefix for workspaces

2.7 Master data

EPLAN distinguishes between two types of master data: **system master data** and **project master data**.

System master data is stored in the directories with the associated user directory that were set during installation. In addition to the user-specific system master data, the original EPLAN system master data is also installed in the *EPLAN original master data directory* (depending on the directory selected during the installation).

It should be noted that on installation, only the original EPLAN system master data is overwritten or updated in the *\EPLAN* directories. In case of a new installation or installation of an update, the user-specific system master data is **not** overwritten or updated. Generally, though, it is possible to synchronize the user's own system master data with the original EPLAN system master data. To do so, in the **UTILITIES/MASTER DATA** menu, you call up the **SYNCHRONIZE SYSTEM MASTER DATA** menu item. In the following dialog, you can then update your own system master data accordingly.

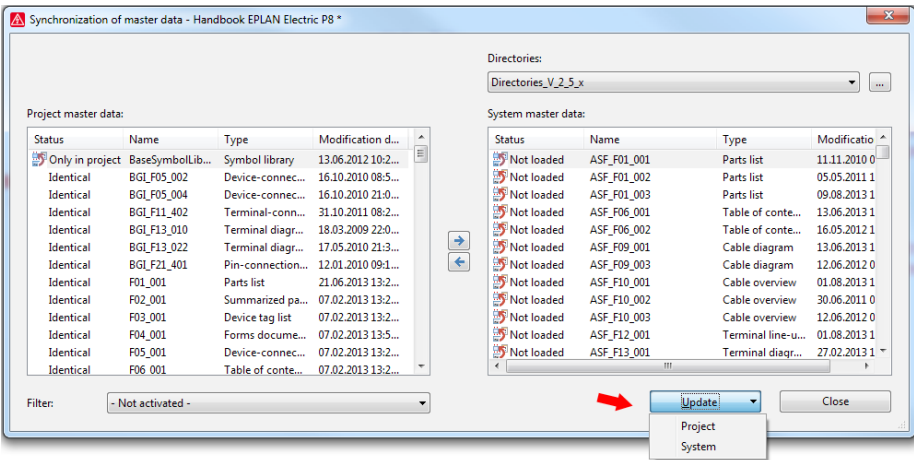


Fig. 2.80 Master data synchronization

If, for example, you have modified the supplied table of contents form *F06_001.f06*, EPLAN would not replace this form in the user-specific master data directories `\\Forms\Company code` with an updated, possibly improved, version. It will only overwrite this form in the `\\Forms\EPLAN` directory.

System master data contains data such as symbol libraries with the associated symbols, function definitions, forms and plot frames.

Project master data is the other type of master data. Project master data, after initial use, is moved from the system master data and stored in the project; after it is stored it is independent of the system master data.

■ 2.8 Operation

EPLAN can be conveniently operated using the mouse. However, to increase your own working speed, EPLAN is very flexible in allowing functions accessed with the mouse to also be accessed via definable keyboard shortcuts. Normally functions can be assigned to a keyboard shortcut if they are also accessible in the main menus, such as Page, Project etc.

To translate text in a Text dialog, you can select the appropriate command with the mouse via **UTILITIES/TRANSLATION**. This function can be assigned a keyboard shortcut because it is a normal menu command. One possibility is the key combination **CTRL + T**.



2.8.1 Using the keyboard

EPLAN allows the user a great deal of freedom in assigning keyboard shortcuts to functions and commands.



NOTE: This is one limitation here: Certain shortcuts are often only permitted in combination with the CTRL and/or ALT keys.

Of course, important standard Windows shortcuts such as **CTRL+ C (COPY)** or **CTRL + V** (paste) should **not** be reassigned if at all possible. It is possible, but not recommended.

Selected shortcuts

Here are a number of recommended keyboard shortcuts that have proven useful in practice.

Copy page from /to	CTRL + SHIFT + P
Edit /Copy format	ALT + F
Edit /Assign format	ALT + A
Utilities /Reports /Generate	CTRL + ALT + G
Find /Synchronize selection	CTRL + SHIFT + Y
Project data /Device navigator	STRG + SHIFT + D
Project data /Update connections	F11
Utilities /Parts /Management	ALT + V
Utilities /Translation /Dictionary	ALT + T

To assign your own shortcut to functions, in the **OPTIONS/SETTINGS** menu, select **USER/MANAGEMENT/SHORTCUT KEYS**.

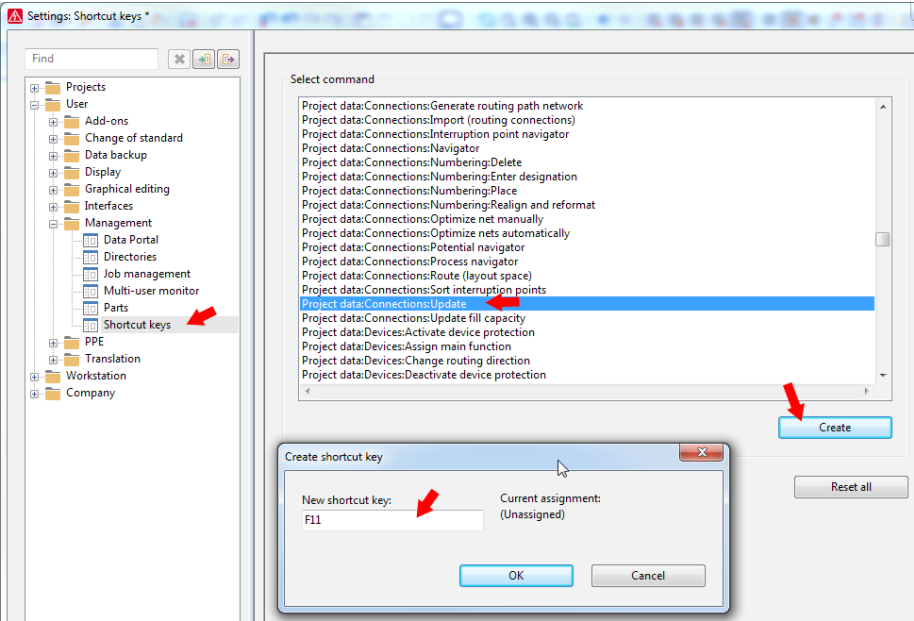


Fig. 2.81 Assigning a shortcut key to a function

In this dialog, you select the desired action and assign it a shortcut key using the **CREATE** button. The shortcut key is applied and fully “operational” as soon as you click **OK**.

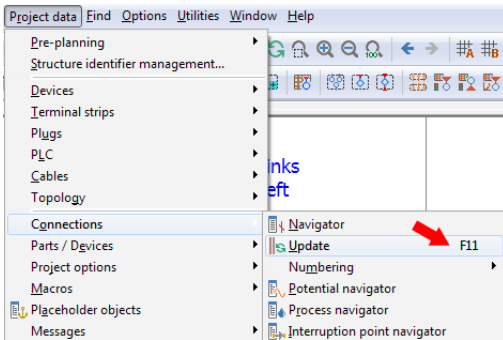


Fig. 2.82

Assigned shortcuts in the Connections menu

2.8.2 Using the mouse

EPLAN can be fully operated using the mouse. EPLAN can be completely operated using the mouse. Admittedly, many menu items are easier to reach with the mouse than with cryptic key combinations that one usually cannot remember.

■ 2.9 User interface – more useful information

EPLAN provides a great deal of support to the user when editing projects, but also for general operation of the program.

A very nice and practical feature is the ability to freely configure the EPLAN user interface. The so-called workspaces are defined depending on the logged-in EPLAN user. This makes it possible to always call up your own user interface configuration when you start up EPLAN on the same computer.

2.9.1 Using workspaces

Put simply, a **workspace** is the dialog layout and the toolbars you want to have when performing certain operations in EPLAN. EPLAN allows you to create separate workspaces for particular areas that contain exactly these desired toolbars, views or dialogs.

The Workspace function is accessed from the graphical editing menu via **VIEW/WORKSPACE**. When you select this function, EPLAN opens the **Workspace** dialog and you can use the selection field to select one.

2.9.1.1 Creating and selecting a workspace

A workspace is customized by placing desired dialogs or toolbars to suit your personal preferences. To permanently save a new workspace, you open the **Workspace** dialog via the **VIEW/WORKSPACE** menu.

A new workspace is created via the  button. EPLAN then opens the **New scheme** dialog. You should enter a name not yet assigned in the *Name* field and a sensible description for the workspace in the *Description* field.

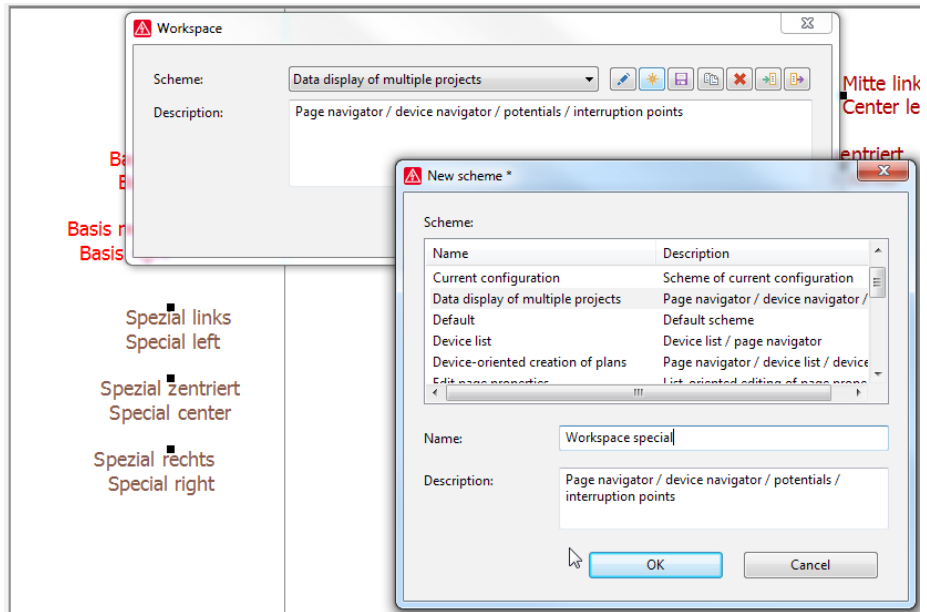


Fig. 2.83 Creating a workspace

When you click **OK**, the workspace is saved and can then be set as the current workspace.

2.9.2 Dialog display

EPLAN provides a lot of data and additional information. EPLAN has a number of different dialog representations in the navigators to ensure information is clearly displayed.

2.9.2.1 Tree view

The **tree view** is one type of representation. Similar to Windows Explorer, all information is displayed in a tree with small symbols.

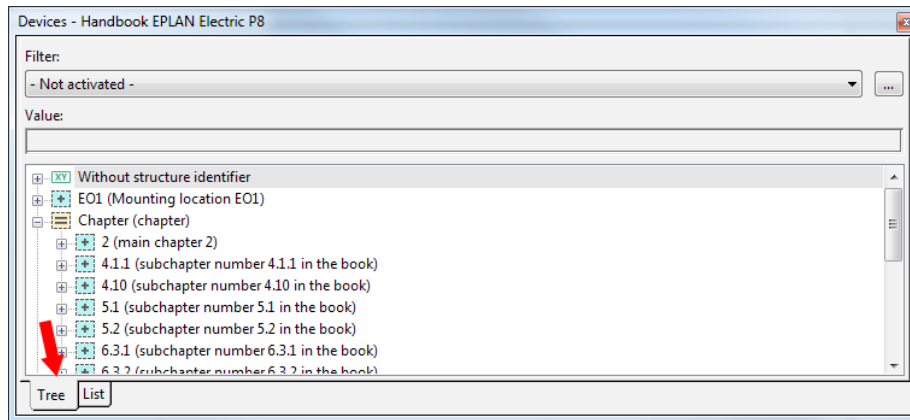


Fig. 2.84
Tree view

This representation type provides a clear overview, making it relatively easy to find objects. In this representation, EPLAN always shows all devices (depending on the function definition). No changes to the form and content of the tree view are possible. EPLAN provides no options for this here.

The display sequence of the devices can be configured under **OPTIONS/SETTINGS/PROJECTS [PROJECT NAME]/DISPLAY/PROJECT STRUCTURE (NAVIGATORS)** and/or **PROJECT STRUCTURE (PAGES)**. This is where you can define how devices should be shown in the tree view, for example, by identifier or by the page prefix.

2.9.2.2 List view

The **list view** is another representation type. This representation type offers many more customization options to suit your working habits. As with the tree view, EPLAN shows all devices here as well (again depending on the navigator that was selected). The cable navigator shows only cables, the terminal navigator shows only devices with the terminal function type, etc.

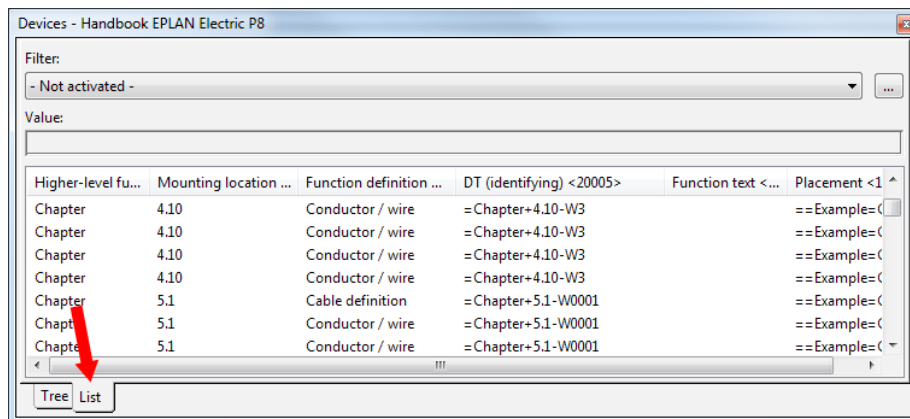


Fig. 2.85
List view

Unlike the tree view, it is possible to add further information to the list view. Right-click to call up the popup menu to select the *Configure representation* function. EPLAN opens the **Configure representation** dialog. You can now select or deselect columns and change their order. Click the OK button to save the settings. You now have a personalized list view.

2.9.2.3 Combined tree and list view (part)

In addition to the previous representations, the *Part master data* navigator also allows a combined display of list and tree view, the **combination** view.

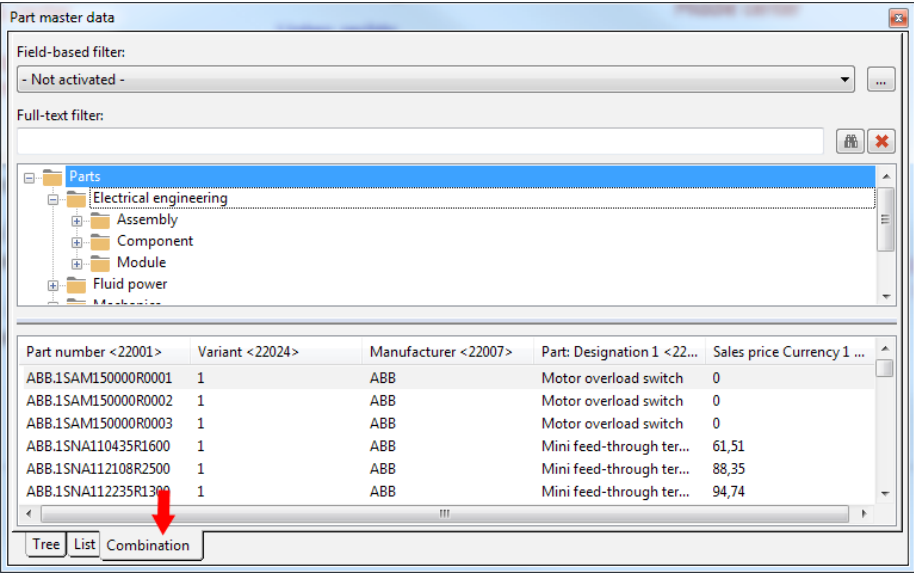


Fig. 2.86 Combined display

The tree view is displayed in the upper area. Here it is possible to preselect, for instance by selecting the *Cable* node. In the lower area, the preselected parts are shown in a clear list with additional specific information.

2.9.2.4 Edit in table

Edit in table is another method of conveniently displaying and editing properties. Edit in table is usually accessed from the tree view by right clicking to access the popup menu. Unlike the previous representations (tree or list view), only the devices and their functions that were previously selected in the navigator are displayed.

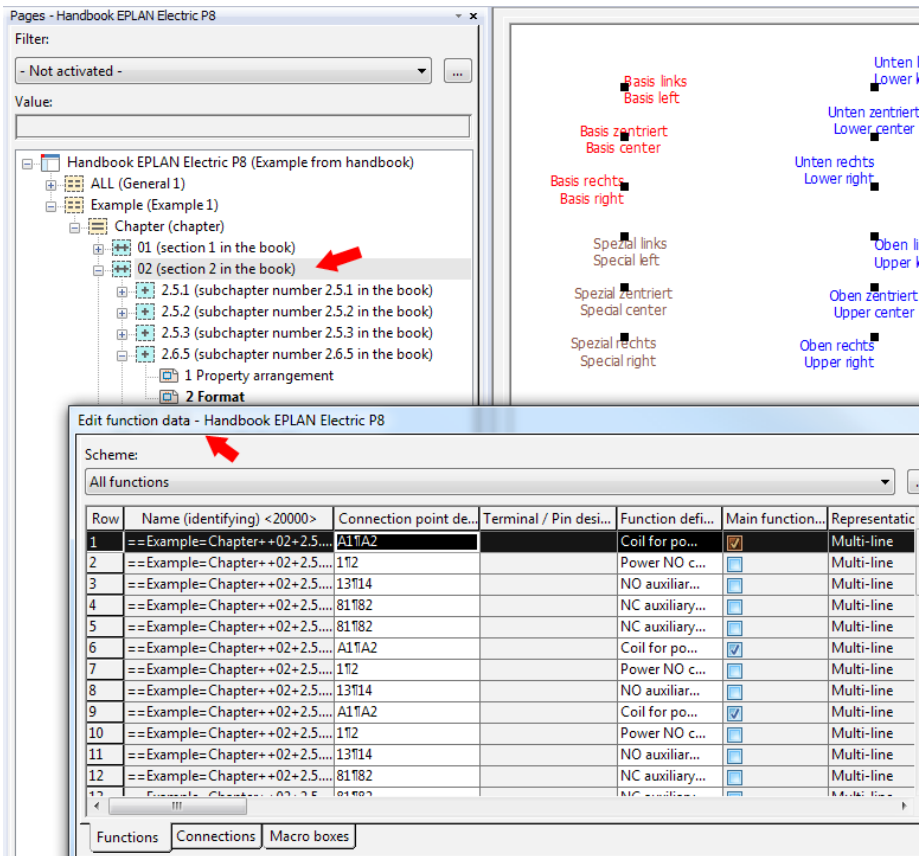


Fig. 2.87 Editing function data in table

The displayed devices can now be edited in a table in the **Edit function data** dialog. All editing functions, such as Copy, Paste and Edit, can be used when editing tables.

3

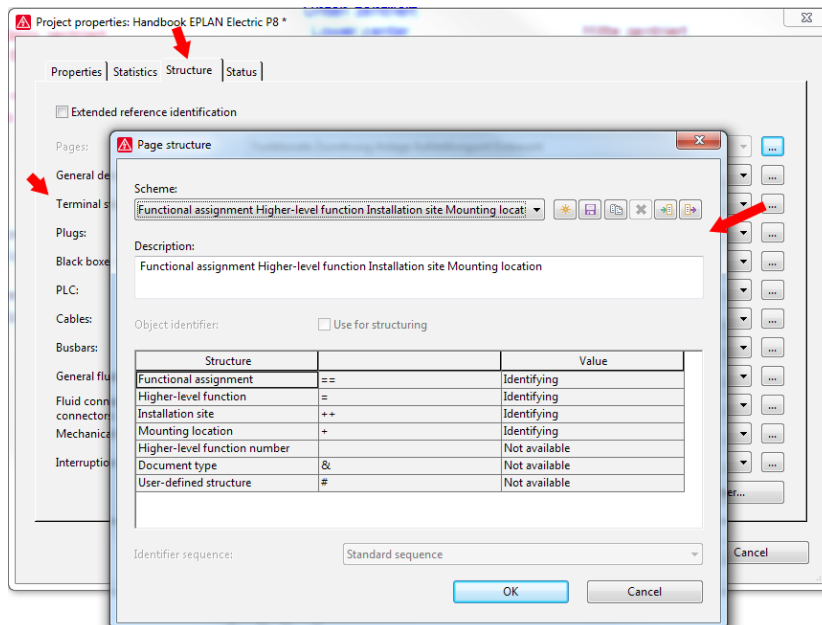
Projects

A project must first be created before you can edit schematics in EPLAN.



NOTE: In EPLAN, the storage location of projects and the names of the individual projects can be freely chosen. The only limitations are the naming conventions of the Windows system being used.

When the page and device structures are defined, a project has all the properties it needs, such as the *device structure* (the structure identifier, e.g. higher-level function or installation site, that is to be used) and, during the time a project is edited, also all the master data, such as forms, parts data, function definitions, symbol libraries, etc., that is necessary for editing projects.



Global structure
of a project

Fig. 3.1
Project properties –
Structure tab